# Chapter 2: Asset Classes and Financial Instruments

**Multiple Choice**

1. A Treasury bill with a face value of $10,000 and 120 days to maturity is selling for $9,780. What are its bank discount yield (BDY) and bond equivalent yield (BEY)?
a. BDY 6.60%; BEY 6.84%
b. BDY 6.84%; BEY 6.60%
c. BDY 6.69%; BEY 6.84%
d. BDY 6.60%; BEY 6.69%

2. For the same Treasury bill, which ordering of the three yield measures is correct?
a. Effective annual yield < bond equivalent yield < bank discount yield
b. Bond equivalent yield < bank discount yield < effective annual yield
c. Bank discount yield < bond equivalent yield < effective annual yield
d. All three are equal because they describe the same bill

3. Which of the following money market instruments is quoted on an interest-bearing (add-on) basis rather than on a discount basis?
a. Negotiable certificate of deposit
b. Treasury bill
c. Commercial paper
d. Banker's acceptance

4. In a repurchase agreement (repo), the dealer:
a. Buys securities from an investor with an agreement to sell them back later at a lower price
b. Sells securities to an investor with an agreement to buy them back the next day at a higher price
c. Lends securities to an investor in exchange for an unsecured promissory note
d. Exchanges dollar deposits for foreign-currency deposits at an agreed forward rate

5. A municipal bond yields 3.2% and a taxable corporate bond of comparable risk yields 4.5%. For an investor in the 32% marginal tax bracket, what is the equivalent taxable yield of the municipal bond, and which bond is better?
a. Equivalent taxable yield 4.71%; the corporate bond
b. Equivalent taxable yield 2.18%; the corporate bond
c. Equivalent taxable yield 4.71%; the municipal bond
d. Equivalent taxable yield 4.22%; the municipal bond

6. A taxable corporate bond yields 7.0% and a comparable-risk municipal bond yields 4.2%. At what marginal tax rate would an investor be indifferent between the two?

a. 28%

b. 35%

c. 40%

d. 60%

7. A Treasury bond with a par value of $1,000 is quoted at an asked price of 103-12. What is the dollar price of the bond?

a. $1,031.20

b. $1,031.88

c. $1,033.75

d. $1,037.50

8. A TIPS bond has a par value of $1,000 and a coupon rate of 2.0%. Inflation over the year is 3.5%, and the principal is adjusted accordingly. What coupon payment does the holder receive for the year?

a. $20.00

b. $20.70

c. $35.00

d. $55.70

9. Which pair of characteristics best describes common stock?

a. Fixed claim and unlimited liability

b. Residual claim and unlimited liability

c. Residual claim and limited liability

d. Fixed claim and limited liability

10. Preferred stock is often called a hybrid of debt and equity because:

a. It can be converted into the firm’s bonds at the holder’s discretion at any time and pays interest instead of dividends

b. It carries full voting rights like common stock and a guaranteed maturity date like a bond

c. It pays a fixed dividend like a perpetual bond with no maturity, yet it is equity that ranks below bondholders and above common stockholders

d. Its dividend is tax-deductible for the issuer, like interest, and rises automatically with the firm’s earnings

11. A foreign company's common share trades at KRW 540,000 in Seoul. One ADR represents 1/10 of a common share, and the exchange rate is KRW 1,350 per U.S. dollar. Ignoring costs, what is the ADR's approximate dollar price?

a. $4.00

b. $40.00

c. $400.00

d. $54,000

12. Consider the three stocks in the table, where Pt is the price and Qt the shares outstanding at time t. Stock C splits two-for-one in the last period (its price falls from 132 to 70 after the split). What is the rate of return on a price-weighted index for the first period (t = 0 to t = 1), and what must the divisor become for the second period so that the split does not change the index level?

| Stock | P0 | Q0 | P1 | Q1 | P2 | Q2 |
|---|---|---|---|---|---|---|
| A | 80 | 150 | 86 | 150 | 89 | 150 |
| B | 45 | 300 | 42 | 300 | 40 | 300 |
| C | 120 | 100 | 132 | 100 | 70 | 200 |

a. Return 3.20%; new divisor 2.24

b. Return 6.12%; new divisor 3.00

c. Return 6.12%; new divisor 2.24

d. Return 6.12%; new divisor 2.38

13. Using the three stocks in the table (stock C splits two-for-one in the last period), what are the first-period returns (t = 0 to t = 1) on an equally weighted index and on a value-weighted index?

| Stock | P0 | Q0 | P1 | Q1 | P2 | Q2 |
|---|---|---|---|---|---|---|
| A | 80 | 150 | 86 | 150 | 89 | 150 |
| B | 45 | 300 | 42 | 300 | 40 | 300 |
| C | 120 | 100 | 132 | 100 | 70 | 200 |

a. Equally weighted 3.61%; value-weighted 3.20%

b. Equally weighted 3.20%; value-weighted 3.61%

c. Equally weighted 3.61%; value-weighted 6.12%

d. Equally weighted 6.12%; value-weighted 3.20%

14. Which statement about an equally weighted index is correct?
a. It corresponds to a buy-and-hold strategy, so no trading is needed as prices change
b. It gives the greatest weight to the highest-priced stocks, as the Dow Jones Industrial Average does
c. It gives the greatest weight to the largest-capitalization firms, as the S&P 500 does
d. Maintaining it requires periodic rebalancing — selling stocks that have risen and buying stocks that have fallen

15. Which statement about a call option is correct?
a. The holder is obligated to buy the asset at expiration regardless of the asset's price
b. The holder has the right but not the obligation to buy the asset at the exercise price, and loses at most the premium paid
c. The holder's loss is unlimited, because the asset price can rise without bound and the holder must deliver at the strike
d. A call is exercised only when the market price falls below the strike price at or before expiration

16. You buy a call option with a strike price of $50 for a premium of $3 per share, and the stock trades at $57 at expiration. You also buy a put option with a strike price of $60 for a premium of $4. What is your profit per share on the call, and at what stock price does the put break even?
a. Call profit $7; put break-even $56
b. Call profit $4; put break-even $64
c. Call profit $10; put break-even $60
d. Call profit $4; put break-even $56

17. Which statement correctly distinguishes a long futures position from a long call position?
a. The futures buyer has the right but not the obligation to take delivery, while the call buyer is obligated to buy
b. Both positions require an up-front premium payment of the same size
c. The futures buyer's maximum loss is limited to the initial margin posted
d. The futures buyer is obligated to settle at the futures price, while the call buyer will exercise only if it is profitable

# Chapter 2: Asset Classes and Financial Instruments — Answers

**1. a**
BDY = [(F − P) ÷ F] × (360 ÷ n) = (220 ÷ 10,000) × (360 ÷ 120) = 6.60%; BEY = [(F − P) ÷ P] × (365 ÷ n) = (220 ÷ 9,780) × (365 ÷ 120) = 6.84%.

**2. c**
BDY divides by face value over 360 days, BEY divides by the lower purchase price over 365 days, and EAY also compounds, so accuracy and size rise BDY → BEY → EAY.

**3. a**
T-bills, commercial paper and bankers' acceptances are sold below face value (discount basis), whereas CDs, repos, federal funds and Eurodollar deposits pay principal plus interest at maturity (add-on).

**4. b**
A repo is a short-term, often overnight, collateralized loan to the dealer; the higher repurchase price is the interest, and in a reverse repo the investor lends cash and holds the collateral.

**5. c**
Equivalent taxable yield = 3.2% ÷ (1 − 0.32) = 4.71%, which exceeds the corporate bond's 4.5%, so the municipal bond leaves more after tax (3.2% versus 4.5% × 0.68 = 3.06%).

**6. c**
Set after-tax yields equal: 7.0% × (1 − t) = 4.2%, so t = 1 − 4.2 ÷ 7.0 = 40%; investors above this bracket prefer the muni, those below prefer the taxable bond.

**7. c**
The digits after the dash are 32nds of a point: 103 + 12/32 = 103.375% of par, i.e. $1,033.75 per $1,000.

**8. b**
TIPS pay the coupon on the inflation-adjusted principal: $1,000 × 1.035 = $1,035, and $1,035 × 2.0% = $20.70.

**9. c**
Shareholders are last in line, receiving only what remains after creditors and preferred holders are paid, and they can lose no more than the amount invested because personal assets are shielded.

**10. c**
Preferred pays a fixed, often cumulative dividend and has priority over common but ranks below debt; unlike bond interest, its dividends are not tax-deductible for the issuer.

**11. b**
ADR price = share price × ratio ÷ exchange rate = 540,000 × 1/10 ÷ 1,350 = $40.00.

**12. c**
Price-weighted return = (86 + 42 + 132) ÷ (80 + 45 + 120) − 1 = 260 ÷ 245 − 1 = 6.12%; just before the split the index is (86 + 42 + 132) ÷ 3 = 86.67, and after it the sum is 86 + 42 + 66 = 194, so the divisor must be d = 194 ÷ 86.67 = 2.24.

**13. a**
Equally weighted = average of the three returns = (7.50% − 6.67% + 10.00%) ÷ 3 = 3.61%; value-weighted = total market value growth = (86×150 + 42×300 + 132×100) ÷ (80×150 + 45×300 + 120×100) − 1 = 38,700 ÷ 37,500 − 1 = 3.20%.

**14. d**
An equally weighted index averages stock returns, i.e. an equal dollar amount in each stock; prices drift the weights apart, so the investor must keep trading back to equal dollars.

**15. b**
The buyer pays a premium for a right: the loss is capped at the premium and the gain is open-ended, and the call is exercised only when the market price is above the strike.

**16. d**
Call profit = max(57 − 50, 0) − 3 = $4; a put pays K − $S_T$, so the premium is recovered when $S_T$ = K − premium = 60 − 4 = $56.

**17. d**
An option is a right, bought for a premium, with a capped loss; a futures contract is an obligation with no premium (margin only), marked to market daily, so a move against you must still be settled.

# Chapter 3: Securities Markets

**Multiple Choice**

1. Which of the following transactions takes place in the primary market?
a. An investor buys 100 shares of Samsung Electronics from another investor on the exchange
b. A mutual fund sells its holdings of Toyota stock to a pension fund
c. A firm sells newly issued shares to the public in an initial public offering
d. A dealer buys shares into inventory from a customer

2. In a firm commitment underwriting, the investment bank:
a. Acts only as an agent and returns any unsold shares to the issuer
b. Guarantees the issuing firm a minimum share price for one year after the offering
c. Buys the entire issue from the firm and bears the placement risk of reselling it to the public
d. Lends the issuing firm the offering proceeds, to be repaid out of future earnings

3. A firm sells 200,000 shares in an initial public offering at an offer price of $25. The underwriter’s explicit fees are $150,000, and the share price closes at $28 on the first day of trading. What is the best estimate of the total cost of the equity issue to the firm?
a. $150,000
b. $600,000
c. $750,000
d. $4,850,000

4. The sell side of a limit-order book for a stock shows the orders in the table. You submit a market order to buy 700 shares. What total amount will you pay, ignoring commissions?

| Ask price | Shares offered |
|---|---|
| $31.40 | 300 |
| $31.45 | 500 |
| $31.55 | 600 |

a. $21,980
b. $22,000
c. $22,015
d. $22,085

5. Which statement correctly contrasts a market order with a limit order?
a. A market order guarantees a price but not execution; a limit order guarantees execution but not price
b. A market order executes immediately at the best available price; a limit order executes only at the specified price or better, and may never execute
c. Both orders execute immediately at the quoted price, but a limit order pays a lower commission
d. A limit order can only be used to buy, while a market order can be used to buy or sell and is the only order allowed during trading hours

6. You own a stock at $120 and place a stop-loss order at $108. Overnight, bad news makes the stock open the next morning at $102. What happens?
a. The shares are sold at exactly $108, because a stop-loss order guarantees the stop price
b. The order is never triggered, because the price skipped over $108 without trading at it
c. The stop is triggered and a market order sells the shares near the $102 opening price
d. The order becomes a limit order at $108, which cannot be filled at the lower opening price

7. Which statement correctly contrasts the NYSE and Nasdaq?
a. The NYSE is an auction, order-driven market with a central limit-order book, while Nasdaq is a dealer, quote-driven market with competing market makers
b. The NYSE is a dealer market with competing market makers, while Nasdaq is an auction market with a central limit-order book
c. Both are pure dealer markets with no limit-order book
d. Nasdaq has a physical trading floor, while the NYSE has no human oversight at all

8. You have $15,000 of cash and the initial margin requirement is 40%. What is the maximum value of stock you can buy, and how much do you borrow? (Ignore commissions.)
a. Buy $37,500; borrow $15,000
b. Buy $37,500; borrow $22,500
c. Buy $30,000; borrow $15,000
d. Buy $25,000; borrow $10,000

9. An investor buys 800 shares at $25 per share using the maximum allowable initial margin of 50%, so the margin loan is $10,000. The maintenance margin is 35%. At what share price will the investor receive a margin call?
a. $8.75
b. $12.50
c. $16.25
d. $19.23

10. An investor buys 300 shares at $60 per share, borrowing 40% of the purchase price at an interest rate of 7% per year. One year later the shares are sold at $66. The stock pays no dividends and commissions are ignored. What is the rate of return on the investor’s equity?

a. 9.67%

b. 10.00%

c. 12.00%

d. 16.67%

11. Which statement about buying stock on margin is correct?

a. Margin reduces the volatility of the investor’s return because the loan is a fixed obligation

b. Margin magnifies both gains and losses on the investor’s equity, and interest on the loan accrues whatever the stock does

c. The investor’s maximum loss on a margined position equals the loss on an unmargined position

d. The initial margin requirement is set by each individual broker and cannot be changed by regulators

12. Which statement about the cash flows and obligations in a short sale is correct?

a. The proceeds are held in the account as collateral, additional margin must be posted, and the short seller must reimburse any dividends the stock pays

b. The short seller receives the sale proceeds in cash and may withdraw them immediately to invest elsewhere, with no further obligation

c. The short seller is entitled to receive any dividends paid on the borrowed shares, because the seller is the owner of record

d. The position requires no margin because the shares were sold rather than bought, so no capital is at risk

13. An investor sells short 200 shares at $40 per share and deposits the required initial margin of 50% in cash. The maintenance margin is 35%. At what share price will the investor receive a margin call?

a. $39.00

b. $44.44

c. $54.00

d. $60.00

14. An investor sells short 500 shares at $60 per share and deposits an initial margin of 50% of the short value. One year later the investor covers at $54, after the stock paid a dividend of $1.50 per share. Treating the initial margin as the amount invested, what is the rate of return on the short position?

a. 7.50%

b. 15.00%

c. 20.00%

d. 25.00%

15. Which statement about U.S. securities regulation is correct?

a. The Securities Exchange Act of 1934 tightened governance after the 2002 scandals, and the CFTC regulates futures

b. The Sarbanes–Oxley Act of 2002 tightened auditing and governance standards after accounting scandals, and the CFTC regulates futures

c. The Sarbanes–Oxley Act of 2002 tightened governance, and FINRA is the government agency that regulates futures

d. The Securities Act of 1933 tightened governance after the 2002 scandals, and the SEC regulates futures

# Chapter 3: Securities Markets — Answers

**1. c**
The primary market is where new securities are issued and the issuing firm receives the proceeds; every later trade among investors is a secondary-market trade that brings the firm no new money.

**2. c**
The underwriter sets an offer price from book-building demand and, under a firm commitment, bears the placement risk; its compensation is the spread between the price it pays and the price it charges.

**3. c**
Cost = explicit fee + money left on the table = \$150,000 + (28 − 25) × 200,000 = \$750,000.

**4. b**
A market buy fills against the lowest asks first: 300 × \$31.40 + 400 × \$31.45 = \$22,000 (an average of \$31.43).

**5. b**
A market order buys certainty of execution but not of price; a limit order buys certainty of price but not of execution, and you rarely get both.

**6. c**
A stop-loss guarantees execution, not price, so after a gap down it fills near the opening price; a stop-limit order, by contrast, would guarantee the price but might not fill at all.

**7. a**
The NYSE uses a central limit-order book with price–time priority (with human oversight at the open and close); Nasdaq has no trading floor and liquidity comes from multiple dealers posting quotes.

**8. b**
Buying power = cash ÷ initial margin = \$15,000 ÷ 0.40 = \$37,500; the loan is \$37,500 − \$15,000 = \$22,500.

**9. d**
Margin = (800P − 10,000) ÷ 800P = 0.35 → 800P × 0.65 = 10,000, so P = \$19.23; below this price a margin call is issued.

**10. c**
Position = 300 × \$60 = \$18,000; equity = \$10,800, loan = \$7,200; ending equity = 300 × \$66 − \$7,200 × 1.07 = \$12,096, so the return is 12,096 ÷ 10,800 − 1 = 12.00% (versus 10.00% unlevered).

**11. b**
The shares are collateral for a fixed loan, so all price variation falls on the investor's equity; leverage scales the return up and down, and Reg T sets initial margin at a minimum of 50% equity.

**12. a**
Short sales must be done in a margin account: the proceeds are not withdrawn, the investor also deposits initial margin, and dividends and borrow fees are costs of staying short.

**13. b**
Collateral = proceeds \$8,000 + margin \$4,000 = \$12,000; margin = (12,000 − 200P) ÷ 200P = 0.35, so P = 12,000 ÷ (200 × 1.35) = \$44.44; for a short position the call comes when the price rises.

**14. b**
Profit = (60 − 54 − 1.50) × 500 = \$2,250 (the dividend must be reimbursed); the investment is the margin of \$15,000, so the return is 2,250 ÷ 15,000 = 15.00%.

**15. b**
Sarbanes–Oxley (2002) responded to the accounting scandals; the SEC oversees securities and security options, the CFTC oversees futures, and FINRA is the industry self-regulatory body.

# Chapter 4: Mutual Funds and Other Investment Companies

**Multiple Choice**

1. Which of the following is NOT one of the four core services an investment company provides to small investors?
a. Diversification and divisibility
b. Lower transaction costs from trading in large blocks
c. Record keeping and administration
d. A guaranteed return above the market index

2. A fund holds 120,000 shares of A at $52, 200,000 shares of B at $18, 90,000 shares of C at $64, and 350,000 shares of D at $9.50. Liabilities are $0.9 million and there are 2.2 million fund shares outstanding. What is the NAV per share?
a. $6.68
b. $8.19
c. $8.60
d. $9.01

3. An open-end fund has a NAV of $32.00. An investor buys $1.6 million of the fund. How many new shares are issued, and what is NAV afterward?
a. 50,000 shares; NAV rises to $32.80
b. 1,600,000 shares; NAV stays at $32.00
c. No new shares; NAV rises
d. 50,000 shares; NAV stays at $32.00

4. You place an order to buy shares of an open-end mutual fund at 10:30 a.m. At what price is the order filled?
a. At the last NAV published before 10:30 a.m.
b. At the market price quoted on the exchange at 10:30 a.m.
c. At any limit price you specify, if the market reaches it
d. At the NAV struck at the 4:00 p.m. close that day

5. Which statement about closed-end funds is correct?
a. Shares outstanding are fixed after the IPO, and the market price can differ from NAV
b. Investors redeem shares directly with the fund at NAV whenever they wish
c. Shares outstanding change daily as investors buy new shares from, and redeem them to, the fund
d. The market price is required by law to equal NAV at the close of each trading day

6. You buy a closed-end fund at $21.00 when its NAV is $24.00. One year later NAV is still $24.00, the price is $22.50, and the fund distributed $1.05 per share. What is your rate of return?
a. 4.38%
b. 7.14%
c. 11.33%
d. 12.14%

7. Which feature describes a unit investment trust (UIT)?
a. An actively managed portfolio overseen by a board of directors
b. A private pool open only to accredited investors
c. A fixed, unmanaged portfolio that terminates on a set date
d. A fund that trades continuously on an exchange at NAV

8. Compared with a mutual fund, a hedge fund typically:
a. Offers daily redemption at NAV to the general public, like an open-end mutual fund
b. Faces tighter limits on leverage and short sales because it is regulated more heavily
c. Is open only to accredited or institutional investors and is largely exempt from the Investment Company Act
d. Charges a lower fee than a mutual fund because it is less regulated and pays no incentive fee

9. Which statement about 12b-1 fees is correct?
a. They are one-time charges paid when shares are purchased, and are deducted from the amount invested
b. They are annual distribution fees of up to 1% of assets and work like a sales load spread over time
c. They are paid to portfolio managers as a performance bonus when the fund beats its benchmark
d. They are charged only on redemption within the first five years, and decline by one point annually

10. A fund reports a one-year return of 9%. For an investor who bought Class A shares with a 4% front-end load, the reported return:
a. Understates her return, because loads are refunded after one year
b. Equals her return, because loads are included in the expense ratio
c. Equals her return, because loads are deducted from NAV
d. Overstates her return, because published returns are computed before loads

11. You invest $30,000 in a fund with a 6% front-end load. How much is actually invested in the fund, and what return must the fund earn for you to get back to $30,000?
a. $28,200 invested; 6.00% return needed
b. $28,302 invested; 6.00% return needed
c. $31,800 invested; 5.66% return needed
d. $28,200 invested; 6.38% return needed

12. You invest $20,000 in Class A shares with a 4% front-end load. The fund’s reported return for the year is 12.0%. What is your return?

a. 7.52%

b. 8.00%

c. 11.54%

d. 12.00%

13. A fund charges a CDSC of 5%, 4%, 3%, 2% and 1% in years 1–5 and 0% thereafter. You redeem shares worth $22,000 during year 4. If the charge applies to the redemption value, what is it?

a. $220

b. $440

c. $660

d. $880

14. A fund’s NAV rises from $24.60 to $25.90 during the year. It pays income distributions of $0.45 and capital gain distributions of $0.95 per share, and its expense ratio is 1.25%. What are its net return and approximate gross return?

a. Net 10.98%; gross 12.23%

b. Net 5.28%; gross 6.53%

c. Net 10.42%; gross 11.67%

d. Net 10.98%; gross 9.73%

15. You invest $10,000 in a fund with a gross return of 8% a year. The two share classes are shown in the table. Which class leaves you with more after 1 year, and which after 10 years?

| Feature | Class A | Class C |
|---|---|---|
| Front-end load | 4.5% | none |
| Expense ratio | 0.75% | 1.45% |
| CDSC | none | 1% if redeemed in year 1 |

a. 1 year: Class A ($10,242); 10 years: Class A ($19,230)

b. 1 year: Class C ($10,548); 10 years: Class C ($18,860)

c. 1 year: Class C ($10,548); 10 years: Class A ($19,230)

d. 1 year: Class A ($10,725); 10 years: Class C ($18,860)

16. Why do ETF holders generally avoid the capital gains distributions that mutual fund holders receive when other investors redeem?
a. ETFs are exempt from capital gains tax by law, so no gains are ever distributed
b. ETFs hold only bonds, which by their nature do not produce capital gains
c. ETF sellers trade with other investors on the exchange, so the fund does not have to sell stock to raise cash
d. ETF managers never sell securities for any reason, not even to rebalance the index

17. An ETF trades at $75.60 while its underlying basket is worth $75.00 per ETF share. Creation units are 20,000 shares. Ignoring trading costs, what is the AP's profit from one creation, and what does the AP do?
a. $12,000 — buy ETF shares, redeem them, sell the basket
b. $12,000 — buy the basket, create ETF shares, sell them
c. $1,200 — buy the basket, create ETF shares, sell them
d. $24,000 — buy ETF shares, redeem them, sell the basket

18. An index moves +8%, −8%, +8%, −8% on four consecutive days, starting at 100. A 2× daily leveraged ETF also starts at 100. What are the ending values?
a. Index 98.72; 2× ETF 94.95
b. Index 98.72; 2× ETF 97.45
c. Index 100.00; 2× ETF 100.00
d. Index 98.72; 2× ETF 98.72

19. You own a stock at $40 and sell a call with a $44 strike for a $1.20 premium. What is your profit per share on the covered call if the stock ends at $50, and if it ends at $35?
a. $10.00 at $50; −$5.00 at $35
b. $11.20 at $50; $1.20 at $35
c. $5.20 at $50; −$3.80 at $35
d. $5.20 at $50; −$6.20 at $35

20. Why must the average actively managed dollar underperform the market after costs?
a. Active managers are, on average, less skilled than the managers who build index funds
b. Before costs, active investors as a group hold the market, so their average gross return equals the market return
c. Active funds hold more cash than the market does, which drags returns in rising years
d. Regulations force active funds to hold riskier stocks than the index contains

21. Of 164 U.S. large-cap funds in the top quartile in 2021, 20.1% were still in the top quartile in 2022 and 0.0% in 2023, while pure chance would give 6.25%. What does this suggest?

a. Ranking funds by past returns is a good way to choose a fund

b. Past top performance is not a reliable signal of future top performance

c. Top funds deliberately take less risk after a good year

d. The result is caused by survivorship bias inflating the winners

# Chapter 4: Mutual Funds and Other Investment Companies — Answers

**1. d**
The four services are record keeping and administration, diversification and divisibility, professional management and lower transaction costs; whether managers succeed on average is an empirical question.

**2. b**
Assets = 120,000×52 + 200,000×18 + 90,000×64 + 350,000×9.5 = $18,925,000; NAV = (18,925,000 − 900,000) ÷ 2,200,000 = $8.19.

**3. d**
New shares = $1,600,000 ÷ $32.00 = 50,000; assets and shares rise in proportion, so NAV stays at $32.00.

**4. d**
Open-end funds transact only at NAV, struck once daily at 4:00 p.m. ET, so every order that day gets the same price and you do not know it when you order; intraday prices and limit orders belong to ETFs and closed-end funds.

**5. a**
A closed-end fund neither issues nor redeems shares after its IPO; investors trade with each other on an exchange, so supply and demand set a price that may be at a premium or discount to NAV.

**6. d**
Return = (22.50 − 21.00 + 1.05) ÷ 21.00 = 12.14%, measured on the market price actually paid.

**7. c**
A sponsor deposits a fixed portfolio into a trust and sells redeemable units; with essentially no trading afterwards, management fees are very low and the trust is self-liquidating.

**8. c**
Hedge funds are private pools that disclose little, may use short sales, derivatives and heavy leverage, impose lock-up periods, and charge about “2 and 20”.

**9. b**
Named for SEC Rule 12b-1, this annual charge (maximum 1% of assets) pays for marketing and adviser compensation; it is included in the expense ratio but economically it is a load paid in instalments.

**10. d**
Reported returns are computed from NAV, which is net of expenses but not of loads, so a load-fund investor earns less than the published figure.

**11. d**
The load is a percentage of the gross amount paid: $30,000 × (1 − 0.06) = $28,200; to recover $30,000 the fund must earn 30,000 ÷ 28,200 − 1 = 6.38%, because the load is 6% of the gross but 6.38% of the net amount invested.

**12. a**
Amount invested = 20,000 × 0.96 = $19,200; end value = 19,200 × 1.12 = $21,504; return = 21,504 ÷ 20,000 − 1 = 7.52%.

**13. b**
The year-4 charge is 2%: 2% × $22,000 = $440.

**14. a**
Net return = (25.90 − 24.60 + 0.45 + 0.95) ÷ 24.60 = 10.98%, since distributions must be added back and the denominator is the beginning NAV; gross return ≈ net return + expense ratio = 10.98% + 1.25% = 12.23%.

**15. c**
Net growth: A = 8% − 0.75% = 7.25%, C = 8% − 1.45% = 6.55%; after 1 year A = 10,000 × 0.955 × 1.0725 = $10,242 and C = 10,000 × 0.99 × 1.0655 = $10,548 (the 1% CDSC applies in year 1); after 10 years A = 10,000 × 0.955 × $1.0725^{10}$ = $19,230 and C = 10,000 × $1.0655^{10}$ = $18,860.

**16. c**
In a mutual fund a redemption forces the fund to sell stock and realize gains that land on holders who sold nothing, whereas an ETF seller simply finds a buyer on the exchange.

**17. b**
The ETF is rich relative to the basket, so the AP buys the basket and creates ETF shares to sell; profit = (75.60 − 75.00) × 20,000 = $12,000.

**18. a**
Index: 100 × 1.08 × 0.92 × 1.08 × 0.92 = 98.72 (−1.28%); 2× ETF: 100 × 1.16 × 0.84 × 1.16 × 0.84 = 94.95 (−5.05%), about four times the index’s loss rather than twice.

**19. c**
At $50 the stock is called away at $44: profit = (44 − 40) + 1.20 = $5.20, so the upside above $44 was sold for the premium; at $35 the call expires worthless: profit = (35 − 40) + 1.20 = −$3.80, because the premium only softens the loss by $1.20 and is not protection.

**20. b**
Passive investors hold the market in proportion, so what remains for active investors is also the market; subtract their higher fees and the average must lag, though this says nothing about any single fund.

**21. b**
If skill drove rankings, winners would repeat more often than chance; the evidence shows no persistence, so past winners are not a dependable guide.

# Chapter 5: Risk, Return, and the Historical Record

**Multiple Choice**

1. You buy a share at KRW 52,000. One year later it trades at KRW 57,200, and you received a cash dividend of KRW 1,300. What is the holding-period return?
a. 2.50%
b. 10.00%
c. 11.36%
d. 12.50%

2. A fund returns +30%, −25%, +15% and −10% in four consecutive years. What are its arithmetic and geometric average returns?
a. Arithmetic 2.50%, geometric 2.50%
b. Arithmetic 0.23%, geometric 2.50%
c. Arithmetic 2.50%, geometric 0.91%
d. Arithmetic 2.50%, geometric 0.23%

3. At t = 0 you buy one share at KRW 50,000. At t = 1 the share pays a KRW 1,500 dividend and you buy a second share at KRW 56,000. At t = 2 each share pays a KRW 1,500 dividend and you sell both shares at KRW 54,000. What are your dollar-weighted and time-weighted (geometric) returns?

| Time | Share price | Dividend per share | Trade |
|---|---|---|---|
| t = 0 | KRW 50,000 | – | Buy 1 share |
| t = 1 | KRW 56,000 | KRW 1,500 | Buy 1 more share |
| t = 2 | KRW 54,000 | KRW 1,500 | Sell both shares |

a. Dollar-weighted 4.15%; time-weighted 6.76%
b. Dollar-weighted 6.76%; time-weighted 4.15%
c. Dollar-weighted 4.15%; time-weighted 7.05%
d. Dollar-weighted 15.00%; time-weighted 6.76%

4. A deposit quotes 9% compounded monthly. What is the effective annual rate?
a. 0.75%
b. 9.00%
c. 9.31%
d. 9.38%

5. An investment has an effective annual rate of 12%. What is the equivalent continuously compounded annual rate?

a. 11.33%

b. 11.39%

c. 12.00%

d. 12.75%

6. A bond earns a nominal return of 7.5% while inflation is 3.2%. What is the exact real return?

a. 4.08%

b. 4.17%

c. 4.30%

d. 10.94%

7. A portfolio returns 40% in a boom (probability 0.25), 12% in a normal year (0.45), and −18% in a bust (0.30). What are its expected return and standard deviation?

| State | Probability | Return |
|---|---|---|
| Boom | 0.25 | 40% |
| Normal | 0.45 | 12% |
| Bust | 0.30 | −18% |

a. E(r) = 10.00%, σ = 21.49%

b. E(r) = 10.00%, σ = 4.62%

c. E(r) = 11.33%, σ = 23.68%

d. E(r) = 10.00%, σ = 41.09%

8. Two return distributions have the same mean and standard deviation. Distribution N is symmetric, while distribution L has negative skew. Which statement about L is correct?

a. It has a long right tail, so upside surprises are more frequent than in N

b. Its mean, median and mode coincide, as they do in the symmetric distribution N

c. It has a long left tail (mean < median < mode), so the standard deviation understates its downside risk

d. Its standard deviation overstates the risk of large losses, since the left tail is long

9. Compared with a normal distribution having the same mean and standard deviation, a return distribution with positive excess kurtosis has:

a. Thinner tails, so extreme returns are rarer than normality predicts

b. A long left tail only, so the mean lies below the median and mode

c. A taller peak and heavier tails on both sides, so extreme returns occur more often than normality predicts

d. A larger standard deviation by construction, since kurtosis is defined from the variance

10. A portfolio worth ₩5 billion has a daily volatility of 1.4%. Assuming normal returns, what are its one-day 95% VaR and its 10-day 99% VaR? (Use z = 1.645 for 95% and z = 2.326 for 99%.)
a. ₩115.15 million and ₩162.82 million
b. ₩115.15 million and ₩364.14 million
c. ₩115.15 million and ₩1,628.20 million
d. ₩115.15 million and ₩514.88 million

11. A bank reports a one-day 95% VaR of ₩30 million. Which statement is correct?
a. The largest loss the bank can suffer in one day is ₩30 million
b. The bank expects to lose ₩30 million per day on average
c. On 95 days out of 100, the bank will lose exactly ₩30 million
d. On about 5 trading days in 100, the loss is expected to exceed ₩30 million

12. Which statement correctly distinguishes value at risk (VaR) from expected shortfall (ES)?
a. VaR measures the average loss beyond the cut-off, while ES only marks the cut-off at a given confidence level
b. VaR marks the cut-off loss at a confidence level, while ES measures how large the loss is on average once that cut-off is passed
c. VaR and ES always give the same number when returns are normally distributed at any confidence level
d. ES ignores losses beyond the cut-off, which is why regulators prefer it to VaR for capital rules

13. Fund A has E(r) = 14% and σ = 24%; Fund B has E(r) = 9% and σ = 10%. The risk-free rate is 3%. Which fund has the higher Sharpe ratio?
a. Fund B: 0.90 versus 0.58
b. Fund A: 0.46 versus 0.60, because it earns more
c. Fund A: 2.18 versus 1.67
d. Fund B: 0.60 versus 0.46

14. As the investor changes y (the fraction of wealth in risky portfolio P, with the rest in the risk-free asset), what happens to the Sharpe ratio of the complete portfolio?
a. It is unchanged for every y
b. It rises as y rises
c. It falls as y rises
d. It rises until y = 1 and then falls

15. Over 1928–2021 the S&P 500 had an arithmetic average return of 11.82% and 3-month T-bills averaged 3.38%. What is the equity risk premium over T-bills?
a. 3.50%
b. 6.60%
c. 8.44%
d. 15.20%

16. Risky portfolio P has E(rP) = 15% and σP = 24%; the risk-free rate is 5%. An investor places y = 0.8 of her wealth in P and the rest in the risk-free asset. What are the expected return, standard deviation and Sharpe ratio of the complete portfolio?

a. E(rC) = 13.00%; σC = 24.00%; Sharpe 0.33

b. E(rC) = 13.00%; σC = 19.20%; Sharpe 0.42

c. E(rC) = 12.00%; σC = 19.20%; Sharpe 0.36

d. E(rC) = 13.00%; σC = 19.20%; Sharpe 0.68

17. Wealth is KRW 30 million, y = 0.6, and risky portfolio P is 55% equities and 45% bonds. How is the money actually allocated?

a. Equities KRW 16.5m, bonds KRW 13.5m, T-bills KRW 0.0m

b. Equities KRW 9.9m, bonds KRW 8.1m, T-bills KRW 12.0m

c. Equities KRW 18.0m, bonds KRW 0.0m, T-bills KRW 12.0m

d. Equities KRW 6.6m, bonds KRW 5.4m, T-bills KRW 18.0m

18. The portfolios in the table are available to investors with utility $U = E(r) - \tfrac{1}{2} \cdot A \cdot \sigma^2$ and A = 2, 5 and 10. Which portfolio does each investor choose?

| Portfolio | E(r) | σ |
|---|---|---|
| Conservative | 6% | 4% |
| Moderate | 10% | 12% |
| Aggressive | 15% | 22% |

a. A = 2: Conservative; A = 5: Moderate; A = 10: Aggressive

b. All three investors choose Aggressive (highest E(r))

c. A = 2: Moderate; A = 5: Moderate; A = 10: Aggressive

d. A = 2: Aggressive; A = 5: Moderate; A = 10: Conservative

19. Portfolio Z has E(r) = 14% and σ = 24%. An investor with A = 4 can instead hold a 3% T-bill. What is Z’s certainty equivalent, and what does she hold?

a. Certainty equivalent 2.48%; she holds the T-bill

b. Certainty equivalent 2.48%; she holds Z because E(r) = 14% is higher

c. Certainty equivalent −9.04%; she holds the T-bill

d. Certainty equivalent 25.52%; she holds Z

20. E(rP) = 12%, σP = 20%, rf = 3% and the investor’s risk aversion is A = 4. What are the optimal risky weight y*, and the resulting E(rC), σC and utility $U = E(r) - \tfrac{1}{2}A\sigma^2$?

a. y* = 1.1250; E(rC) = 13.12%; σC = 22.50%; U = 3.00%

b. y* = 0.5625; E(rC) = 8.06%; σC = 11.25%; U = 5.53%

c. y* = 0.1125; E(rC) = 4.01%; σC = 2.25%; U = 3.91%

d. y* = 0.5625; E(rC) = 8.06%; σC = 11.25%; U = 8.06%

21. Risky portfolio P has E(rP) = 12% and σP = 20%; the risk-free rate is 3%. An investor with A = 2 has optimal weight y* = 1.125 and wealth of KRW 16 million. Assuming she can borrow at the risk-free rate, what does she do?

a. Invests KRW 2.0m in P and holds the rest in T-bills

b. Borrows KRW 18.0m and invests KRW 34.0m in P

c. Borrows KRW 2.0m and invests KRW 18.0m in P

d. Holds KRW 18.0m in P and KRW 2.0m in T-bills

22. Holding everything else fixed, which change would make the optimal risky weight y* smaller?

a. A rise in the risk premium E(rP) − rf

b. A fall in the coefficient of risk aversion A

c. A fall in the variance of the risky portfolio

d. A rise in the variance of the risky portfolio

23. In $U = E(r) - \frac{1}{2} \cdot A \cdot \sigma^2$, which statement about A is correct?

a. A < 0 describes a risk-averse investor

b. A = 0 describes a risk-neutral investor who judges by E(r) alone

c. A larger A means the investor is more tolerant of risk

d. A has no effect when σ is below 20%

24. Risky portfolio P has E(rP) = 15% and σP = 25%; the risk-free rate is 5%. What is the slope of the capital allocation line, and where does it cross σ = 0?

a. Slope 0.60; intercept 5%

b. Slope 0.40; intercept 5%

c. Slope 0.40; intercept 0%

d. Slope 2.50; intercept 15%

25. What is the capital market line (CML)?

a. The line joining the lowest-risk and highest-return securities in the market

b. The capital allocation line of the best-performing active fund over the sample period

c. A line showing the mean and variance of every individual stock in the market

d. The capital allocation line that uses a broad market index as the risky portfolio P

# Chapter 5: Risk, Return, and the Historical Record — Answers

**1. d**
HPR = (57,200 − 52,000 + 1,300) ÷ 52,000 = 12.50%, i.e. capital gain yield 10.00% plus dividend yield 2.50%.

**2. d**
Arithmetic = (30 − 25 + 15 − 10) ÷ 4 = 2.50%; geometric = (1.30 × 0.75 × 1.15 × 0.90)^(1/4) − 1 = 0.23%, always below the arithmetic mean.

**3. a**
Net cash flows: t0 = −50,000; t1 = −56,000 + 1,500 = −54,500; t2 = 2 × 1,500 + 2 × 54,000 = +111,000, whose IRR is the dollar-weighted return, 4.15%. Period returns: year 1 = (56,000 − 50,000 + 1,500) ÷ 50,000 = 15.00%, year 2 = (54,000 − 56,000 + 1,500) ÷ 56,000 = −0.89%, so the time-weighted return = (1.15 × 0.99107)^(1/2) − 1 = 6.76%; the dollar-weighted return is lower because more money was invested in the weaker second year.

**4. d**
EAR = $(1 + 0.09/12)^{12} - 1$ = 9.38%.

**5. a**
APR (continuous) = ln(1 + EAR) = ln(1.12) = 11.33%.

**6. b**
Fisher equation: 1 + R = (1 + r)(1 + i), so r = 1.075 ÷ 1.032 − 1 = 4.17%.

**7. a**
E(r) = 0.25(40) + 0.45(12) + 0.30(−18) = 10.00%; $\sigma^2$ = $0.25(30)^2 + 0.45(2)^2 + 0.30(-28)^2$ = 462, so σ = √462 = 21.49%.

**8. c**
With negative skew the left tail is long and the mean falls below the median and mode; equity returns are typically left-skewed, so σ alone understates their risk.

**9. c**
Fat tails mean extremes arrive far more often than a normal distribution implies (Black Monday 1987 was about a −20σ day), so normality understates the chance of crashes.

**10. d**
VaR = V0 × z × σ × √T. One-day 95%: ₩5,000 million × 1.645 × 0.014 × 1 = ₩115.15 million; 10-day 99%: ₩5,000 million × 2.326 × 0.014 × √10 = ₩514.88 million.

**11. d**
VaR is the loss exceeded with probability 5% over the stated horizon; it is not a maximum loss and says nothing about how large the loss is on those worst days.

**12. b**
VaR says nothing about the size of losses past the cut-off; ES (also CVaR) answers that and is coherent, and Basel FRTB replaced the 99% VaR charge with a 97.5% expected shortfall.

**13. d**
Sharpe = (E(r) − rf) ÷ σ: A = (14 − 3) ÷ 24 = 0.46; B = (9 − 3) ÷ 10 = 0.60, so B gives more reward per unit of volatility.

**14. a**
Both the risk premium and σC scale with y, so their ratio stays equal to the slope of the CAL; this is why choosing P and choosing y can be solved separately (the separation property).

**15. c**
The equity risk premium is 11.82% − 3.38% = 8.44 percentage points.

**16. b**
E(rC) = rf + y[E(rP) − rf] = 5% + 0.8 × 10% = 13.00%; σC = y·σP = 0.8 × 24% = 19.20%; Sharpe = (13.00 − 5) ÷ 19.20 = 0.42, identical to P's 10 ÷ 24, because the risk premium and σC both scale with y.

**17. b**
T-bills = 30m × 0.4 = KRW 12.0m; P = 30m × 0.6 = KRW 18.0m, of which equities = 18.0 × 0.55 = KRW 9.9m and bonds = 18.0 × 0.45 = KRW 8.1m.

**18. d**
Utilities (A = 2 / 5 / 10): Conservative 5.84 / 5.60 / 5.20; Moderate 8.56 / 6.40 / 2.80; Aggressive 10.16 / 2.90 / −9.20, so more risk aversion means a safer choice.

**19. a**
U = 0.14 − ½ × 4 × $0.24^2$ = 2.48%, below the 3% certain return of the T-bill, so she holds the T-bill; a risk-free asset has σ = 0, so its certainty equivalent is its own return.

**20. b**
y* = [E(rP) − rf] ÷ (A·$\sigma P^2$) = 0.09 ÷ (4 × 0.04) = 0.5625; E(rC) = 3% + 0.5625 × 9% = 8.06%; σC = 0.5625 × 20% = 11.25%; U = 0.0806 − ½ × 4 × $0.1125^2$ = 5.53%.

**21. c**
y* = 1.125 > 1: she invests 1.125 × 16m = KRW 18.0m in P, financed by her own 16.0m plus KRW 2.0m borrowed at the risk-free rate.

**22. d**
y* = [E(rP) − rf] ÷ (A·$\sigma P^2$): a bigger reward invites a bigger bet, while more risk aversion or more variance shrinks it.

**23. b**
A > 0 is risk averse (empirically about 2 to 4), A = 0 is risk neutral and A < 0 is risk loving; the penalty ½·A·$\sigma^2$ grows with both A and σ.

**24. b**
Slope = (E(rP) − rf) ÷ σP = (15 − 5) ÷ 25 = 0.40, the Sharpe ratio, and the line starts at rf = 5% where σ = 0.

**25. d**
A passive strategy holds a broad market index without security analysis or market timing; when that index is P, the resulting CAL is called the CML.

# Chapter 6: Optimal Risky Portfolios

**Multiple Choice**

1. Each of n stocks has a standard deviation of 30% and every pair has a correlation of 0.25. An investor holds them with equal weights. What is the portfolio standard deviation when n = 20, and what does it converge to as n grows without limit?
a. 15.00% at n = 20; floor 16.09%
b. 6.71% at n = 20; floor 0.00%
c. 16.09% at n = 20; floor 7.50%
d. 16.09% at n = 20; floor 15.00%

2. Which of the following is an example of firm-specific (diversifiable) risk?
a. The central bank unexpectedly raises interest rates for the whole economy
b. A pandemic lowers the earnings of nearly all listed firms at once
c. A pharmaceutical company announces that its main drug failed its clinical trial
d. The won depreciates sharply against the dollar, hurting importers and exporters alike

3. A stock has a total return variance of 0.0400 ($\sigma$ = 20%), of which 0.0256 ($\sigma$ = 16%) is systematic. What is the standard deviation attributable to firm-specific risk?
a. 1.44%
b. 4.00%
c. 12.00%
d. 25.61%

4. Four equally likely states produce the returns in the table. What are the covariance (in %²) and correlation between the stock and bond returns?

| State | Prob. | Stock return | Bond return |
|---|---|---|---|
| Severe recession | 0.25 | −20% | +11% |
| Mild recession | 0.25 | +4% | +5% |
| Normal growth | 0.25 | +16% | +2% |
| Boom | 0.25 | +30% | +2% |

a. Cov = −64.5; $\rho$ = −0.96
b. Cov = 64.5; $\rho$ = 0.96
c. Cov = −27.0; $\rho$ = −0.40
d. Cov = −64.5; $\rho$ = −2.93

5. A stock fund has E(r) = 11% and σ = 22%; a bond fund has E(r) = 5% and σ = 8%; their correlation is 0.3. What are the expected return and standard deviation of a portfolio with 70% in stocks and 30% in bonds?

a. E(rP) = 9.20%, σP = 17.80%

b. E(rP) = 9.20%, σP = 15.59%

c. E(rP) = 8.00%, σP = 12.78%

d. E(rP) = 9.20%, σP = 16.28%

6. A stock fund (E(r) = 14%, σ = 30%) and a bond fund (E(r) = 6%, σ = 10%) have a correlation of −1. What weight in stocks creates a zero-risk portfolio, and what is its expected return?

a. 75.0% in stocks; E(r) = 12.0%

b. 25.0% in stocks; E(r) = 8.0%

c. 33.3% in stocks; E(r) = 8.7%

d. 50.0% in stocks; E(r) = 10.0%

7. Stock fund σS = 24%, bond fund σB = 8%, correlation 0.1. What is the weight in stocks of the minimum-variance portfolio, and its standard deviation?

a. 10.00% in stocks; σ = 7.81%

b. 92.55% in stocks; σ = 22.28%

c. 7.45% in stocks; σ = 8.00%

d. 7.45% in stocks; σ = 7.79%

8. A portfolio is 100% in a bond fund with σB = 12%. A stock fund has σS = 30%. For which correlation with the bond fund does adding a small amount of stock reduce the portfolio's risk?

a. ρ = 0.25

b. ρ = 0.40

c. ρ = 0.60

d. ρ = 0.90

9. As the correlation between two risky assets decreases while the expected returns and standard deviations stay the same, what happens to the portfolio opportunity set?

a. It becomes a straight line between the two assets

b. It shifts upward, so every mix earns a higher expected return

c. It bows further to the left, so a given expected return can be earned at lower risk

d. It is unchanged, because expected returns are linear in the weights

10. On the opportunity set of two risky assets, which portfolios are efficient?

a. Those on the upward-sloping part starting at the minimum-variance portfolio

b. Those on the downward-sloping part below the minimum-variance portfolio

c. Every portfolio on the curve

d. Only the two single-asset portfolios

11. Stock fund: E(r) = 13%, σ = 24%; bond fund: E(r) = 5%, σ = 9%; correlation 0.25; rf = 2%. What are the weights, expected return, standard deviation and Sharpe ratio of the optimal risky portfolio?

a. Stocks 39.13%, bonds 60.87%; E(rP) = 8.13%, σP = 12.00%, Sharpe 0.511

b. Stocks 4.92%, bonds 95.08%; E(rP) = 5.39%, σP = 8.93%, Sharpe 0.380

c. Stocks 34.02%, bonds 65.98%; E(rP) = 7.72%, σP = 11.23%, Sharpe 0.509

d. Stocks 60.87%, bonds 39.13%; E(rP) = 9.87%, σP = 15.86%, Sharpe 0.496

12. rf = 2%. Which of the candidate risky portfolios in the table gives the steepest capital allocation line?

| Candidate portfolio | w in stocks | E(rP) | σP |
|---|---|---|---|
| All bonds | 0.00% | 5.00% | 9.00% |
| Minimum-variance portfolio | 4.92% | 5.39% | 8.93% |
| 60 / 40 portfolio | 60.00% | 9.80% | 15.69% |
| All stocks | 100.00% | 13.00% | 24.00% |
| Optimal risky portfolio | 39.13% | 8.13% | 12.00% |

a. Minimum-variance portfolio (lowest risk)

b. All stocks (highest expected return)

c. All bonds (simplest low-risk choice)

d. Optimal risky portfolio (highest Sharpe ratio)

13. The optimal risky portfolio has E(rP) = 8.13%, σP = 12.00% with 39.13% in the stock fund; rf = 2%. What fraction of total wealth should an investor with A = 5 hold in the stock fund?

a. 14.81%

b. 33.33%

c. 39.13%

d. 85.19%

14. Two clients with risk aversion A = 3 and A = 8 face the same inputs and can lend and borrow at the same risk-free rate. According to the separation property:

a. They hold the same stock-to-bond ratio inside the risky portfolio but different amounts of T-bills

b. They hold different risky portfolios, with the more risk-averse client holding more bonds inside P

c. They hold the same total allocation to risky assets

d. The client with A = 8 holds no risky assets

15. Which of the following is a reason why the separation property breaks down in practice?

a. All investors face identical taxes and the same constraints

b. Investors agree on the input list and hold the same expectations

c. Human capital is the same for every investor in the economy

d. Borrowing rates exceed lending rates, which kinks the CAL

16. Which statement about the Markowitz efficient frontier of n risky assets is correct?
a. Each individual asset lies on the efficient frontier, since single securities are its building blocks
b. Individual assets lie inside the frontier, and only the branch above the minimum-variance portfolio is efficient
c. The entire frontier is efficient, including the part below the minimum-variance portfolio
d. The tangency portfolio is found by minimizing variance for a zero expected return

17. To run the Markowitz optimization on n = 200 securities, how many estimates are needed in total, and how many of them are covariances?
a. 19,900 in total; 19,900 covariances
b. 20,300 in total; 39,800 covariances
c. 20,300 in total; 19,900 covariances
d. 40,000 in total; 19,900 covariances

18. Korean households hold mostly domestic equities, although Korea is a small share of world market capitalization. Why would adding foreign equities usually reduce risk?
a. Foreign equities always earn higher expected returns than domestic equities
b. Correlations between countries are exactly zero, so foreign risk is fully independent
c. Adding foreign equities removes market risk entirely, because currencies offset equity moves
d. Foreign markets have different industrial structures, so their correlation with domestic stocks is below +1

# Chapter 6: Optimal Risky Portfolios — Answers

**1. d**
Average covariance = $0.25 \times 30^2 = 225$; variance = $(1/20) \times 900 + (1 - 1/20) \times 225 = 258.75$, so $\sigma = 16.09\%$; as $n \to \infty$ the first term vanishes and $\sigma \to \sqrt{225} = 15.00\%$, the systematic risk that diversification cannot remove.

**2. c**
Firm-specific risk arises from events affecting one company or industry and cancels out in a large portfolio; interest rates, pandemics and exchange rates hit every firm at once, so they are market (systematic) risk.

**3. c**
Var(total) = Var(systematic) + Var(firm-specific) because the two parts are uncorrelated, so firm-specific variance = $0.0400 - 0.0256 = 0.0144$ and $\sigma = \sqrt{0.0144} = 12.00\%$.

**4. a**
E(S) = 8% and E(B) = 5%; Cov = Σ p × (dev S) × (dev B) = −64.5; $\sigma S = 18.35\%$ and $\sigma B = 3.67\%$, so $\rho = -64.5 \div (18.35 \times 3.67) = -0.96$; the stock does worst exactly when the bond does best.

**5. d**
$E(rP) = 0.7(11) + 0.3(5) = 9.20\%$; variance = $0.49(484) + 0.09(64) + 2(0.7)(0.3)(52.8) = 265.10$, so $\sigma P = 16.28\%$.

**6. b**
Set $wS\sigma S = wB\sigma B$: $30w = 10(1 - w)$, so $w = 10 \div (30 + 10) = 0.25$; $E(r) = 0.25(14) + 0.75(6) = 8.0\%$, with zero standard deviation.

**7. d**
$Cov = 0.1 \times 24 \times 8 = 19.2$; $wS = (\sigma B^2 - Cov) \div (\sigma S^2 + \sigma B^2 - 2Cov) = (64 - 19.2) \div (576 + 64 - 38.4) = 0.0745$; $\sigma = 7.79\%$, below the bond fund's own 8%.

**8. a**
Adding the riskier asset lowers risk whenever $\rho < \sigma B / \sigma S = 12/30 = 0.40$; $\rho = 0.25$ satisfies this, while $\rho = 0.40$ is exactly the break-even and higher correlations raise risk.

**9. c**
Expected return depends only on the weights, so diversification shows up as a leftward bow; at $\rho = +1$ the set is a straight line, and at $\rho = -1$ it touches the vertical axis.

**10. a**
Points below the minimum-variance portfolio are dominated by points above it with the same risk but a higher expected return, so only the upper branch is efficient.

**11. a**
Cov = 54.0; RS = 11, RB = 3; $wS^* = [11(81) - 3(54.0)] \div [11(81) + 3(576) - 14(54.0)] = 729 \div 1863 = 0.3913$; $E(rP) = 0.3913(13) + 0.6087(5) = 8.13\%$; $\sigma P^2 = 143.93$, so $\sigma P = 12.00\%$; Sharpe = $(8.13 - 2) \div 12.00 = 0.511$.

**12. d**
Sharpe ratios: All bonds 0.333; Minimum-variance portfolio 0.380; 60 / 40 portfolio 0.497; All stocks 0.458; Optimal risky portfolio 0.511; the CAL slope is the Sharpe ratio, and neither the lowest-risk nor the highest-return portfolio maximizes it.

**13. b**
$y^* = [E(rP) - rf] \div (A \cdot \sigma P^2) = 0.0613 \div (5 \times 0.014393) = 0.8519$; stocks = $y^* \times wS^* = 0.8519 \times 0.3913 = 33.33\%$ of total wealth (look-through).

**14. a**
The tangency portfolio is the same for everyone; only $y^*$ differs, so the more risk-averse client simply puts more in T-bills while keeping the identical mix within P.

**15. d**
The other limits listed in the slides are tax differences, human capital (a semiconductor engineer should tilt away from semiconductors), disagreement over the input list, and binding constraints.

**16. b**
The frontier is found by minimizing variance for each target return; single assets are dominated, and the tangency portfolio P* remains the unique Sharpe-maximizing portfolio.

**17. c**
Total = n expected returns + n variances + $n(n - 1)/2$ covariances = 200 + 200 + 19,900 = 20,300 = $n(n + 3)/2$; covariances are symmetric, so only $n(n - 1)/2 = 19{,}900$ are distinct.

**18. d**
The benefit is a covariance argument, not a forecasting argument: as long as cross-border correlations are below +1 the interaction term helps, although correlations have risen over recent decades and currency risk must be handled separately.

# Investment Analysis High-Difficulty Quiz

Multiple Choice · Chapters 3-7 · 11 questions

1. You sell short 100 shares at $50 per share and post the initial margin of 50% in cash. The maintenance margin is 30%. At what share price will you receive a margin call?
   a. $53.85
   b. $57.69
   c. $62.50
   d. $65.00

2. The stock market returns 8% a year before costs. You invest $10,000 for 20 years in either an index fund with a 0.05% expense ratio or an active fund with a 1.00% expense ratio that earns the market return before fees. How much more does the index fund leave you with? (Net return = gross return − expense ratio.)
   a. $9,580
   b. $7,483
   c. $3,720
   d. $1,900

3. You invest $10,000 for one year at a gross portfolio return of 10%. Class A has a 5% front-end load and 0.90% annual expenses. Class C has no front-end load, 1.60% annual expenses, and a 1% CDSC charged on the redemption value if you redeem in year 1. (Net growth = gross return − expense ratio.) What are the ending values after all charges?
   a. Class A $10,364.50 and Class C $10,840.00
   b. Class A $10,910.00 and Class C $10,731.60
   c. Class A $10,450.00 and Class C $10,840.00
   d. Class A $10,364.50 and Class C $10,731.60

4. Class A has a 5% front-end load and 0.90% annual expenses. Class C has no load and 1.60% annual expenses (ignore any CDSC). The gross return is 10% a year. After roughly how many years does Class A overtake Class C?
   a. About 3 years
   b. About 5 years
   c. About 8 years
   d. About 12 years

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## Exhibit A · Share trades and dividends

*Used in questions 5–6*

Trades and dividends (KRW per share)

| Time | Action | Price | Dividend |
|---|---|---|---|
| t = 0 | Buy 1 share | 40,000 | — |
| t = 1 | Buy 1 more share | 44,000 | 1,000 |
| t = 2 | Sell 2 shares | 46,000 | 1,000 |

5. [Exhibit A] Using the share trades in Exhibit A, what is your dollar-weighted return (IRR) per year? Dividends are paid per share held just before each trade.

a. 9.62%
b. 9.66%
c. 6.82%
d. 8.70%

6. [Exhibit A] Using the share trades in Exhibit A, what is your time-weighted (geometric) return per year? Dividends are paid per share held just before each trade.

a. 9.62%
b. 9.66%
c. 8.70%
d. 12.50%

7. A preferred stock (par value $100) pays an 8% annual dividend rate, paid quarterly. The first dividend will be paid 3 years from today (at the end of quarter 12). The required return is 10% per year, compounded monthly. What is the price today?

a. $58.85
b. $60.33
c. $60.97
d. $79.34

8. Mirae Logistics paid a dividend of $1.35 per share four years ago and has just paid $1.92 per share. Assume dividends keep growing forever at the same average annual rate as over the past four years. The required return is 13%. What is the price today?

a. $36.13
b. $50.59
c. $55.25
d. $86.84

9. A stock pays no dividend in years 1 and 2 and pays $1.80 at the end of year 3. Dividends then grow 20% per year in years 4, 5 and 6, and 5% per year forever after year 6. The required return is 14%. What is the price today?

a. $21.79
b. $21.00
c. $24.84
d. $19.76

10. Gyeongbok Software has just paid a dividend of $2.50 per share. Dividends will grow 18% per year for the next 6 years and 5% per year forever after that. The required return is 14%. Use the two-stage growth model. What is the price today?

a. $32.78
b. $35.87
c. $51.12
d. $52.83

11. Namsan Electronics has just paid a dividend of $1.00 per share and reported EPS of $3.00 for the year just ended. Dividends grow 12% per year for the next 5 years, and EPS grows 10% per year. At the end of year 5 the stock is sold at a forward P/E of 14 (price divided by the next year's EPS). The required return is 12%. What is the price today?

a. $42.22
b. $42.70
c. $43.38
d. $47.22

# Investment Analysis High-Difficulty Quiz — Answers

**1. b**
Assets = 5,000 + 2,500 = $7,500 (sale proceeds + 50% margin)
Liability = 100P (value of the shares owed)
Margin = (7,500 − 100P) / 100P = 0.30
7,500 = 130P, so P = 7,500 / 130 = $57.69
BA II Plus: 7500 ÷ 130 = 57.69

**2. b**
Index net return = 8% − 0.05% = 7.95%
Active net return = 8% − 1.00% = 7.00%
Index FV = $10{,}000 \times (1.0795)^{20} = \$46{,}180$
Active FV = $10{,}000 \times (1.07)^{20} = \$38{,}697$
Difference = 46,180 − 38,697 = $7,483
BA II Plus: 20 N, 7.95 I/Y, −10000 PV, 0 PMT, CPT FV = 46,180
BA II Plus: 20 N, 7 I/Y, −10000 PV, 0 PMT, CPT FV = 38,697

**3. d**
Class A growth = 10% − 0.90% = 9.10%
Class C growth = 10% − 1.60% = 8.40%
Class A = $10{,}000 \times 0.95 \times 1.091 = \$10{,}364.50$ (5% load taken at purchase)
Class C = $10{,}000 \times 1.084 \times 0.99 = \$10{,}731.60$ (1% CDSC on redemption)
Class C is ahead after one year
BA II Plus: 9500 × 1.091 = 10,364.50
BA II Plus: 10840 × 0.99 = 10,731.60

**4. c**
Set $0.95 \times (1.091)^n = (1.084)^n$
$n = \ln(1 / 0.95) / \ln(1.091 / 1.084)$
$\ln(1 / 0.95) = 0.05129$
$\ln(1.091 / 1.084) = 0.006437$
$n = 0.05129 / 0.006437 = 7.97$ years
Check at n = 8: $0.95 \times (1.091)^8 = 1.9069 > (1.084)^8 = 1.9065$
BA II Plus: 1 ÷ 0.95 = LN → 0.05129
BA II Plus: 1.091 ÷ 1.084 = LN → 0.006437
then 0.05129 ÷ 0.006437 = 7.97

**5. d** [Exhibit A]
$CF_0 = -40{,}000$ (buy 1 share)
$CF_1 = -44{,}000 + 1{,}000 = -43{,}000$ (buy a second share, receive the dividend on 1 share)
$CF_2 = 2 \times 1{,}000 + 2 \times 46{,}000 = +94{,}000$ (dividends on 2 shares, sell 2 shares)
Equation: $0 = -40{,}000 - 43{,}000 / (1 + r) + 94{,}000 / (1 + r)^2$
r = 8.70%
BA II Plus: CF, CF0 = −40000, C01 = −43000, F01 = 1, C02 = 94000, F02 = 1, IRR CPT = 8.70

**6. a** [Exhibit A]
Year 1 return = (44,000 − 40,000 + 1,000) / 40,000 = 12.50%
Year 2 return = (46,000 − 44,000 + 1,000) / 44,000 = 6.82% (per share
base is the t = 1 price)
TWR = $[(1.1250)(1.0682)]^{1/2} - 1 = 9.62\%$
BA II Plus: 1.125 × 1.068182 = 1.201705
press the square-root key → 1.09623, then − 1 = 0.0962 = 9.62%

**7. b**
D = 100 × 0.08 / 4 = $2.00 per quarter
$R = (1 + 0.10/12)^3 - 1 = 2.5209\%$ per quarter
$P_{11} = D / R = 2.00 / 0.025209 = \$79.34$ (one quarter before the first dividend)
$P_0 = 79.34 / (1.025209)^{11} = \$60.33$
BA II Plus: 1 + 0.10 ÷ 12 = $y^x$ 3 = 1.025209, so R = 2.5209%
BA II Plus: 11 N, 2.5209 I/Y, 0 PMT, −79.34 FV, CPT PV = 60.33

**8. c**
$g = (1.92 / 1.35)^{1/4} - 1 = 9.2048\%$ (four growth periods, not five)
$D_1 = 1.92 \times 1.092048 = \$2.0967$
$P_0 = D_1 / (R - g) = 2.0967 / (0.13 - 0.092048) = \$55.25$
BA II Plus: 4 N, −1.35 PV, 0 PMT, 1.92 FV, CPT I/Y = 9.2048
BA II Plus: 1.92 × 1.092048 ÷ 0.037952 = 55.25

**9. a**
$D_4 = 1.80 \times 1.20 = 2.160$
$D_5 = 2.160 \times 1.20 = 2.592$
$D_6 = 2.592 \times 1.20 = 3.1104$
$P_6 = D_7 / (R - g) = 3.1104 \times 1.05 / (0.14 - 0.05) = \$36.288$ (built from $D_7$, added to $D_6$ in year 6)
$P_0 = 1.80/(1.14)^3 + 2.160/(1.14)^4 + 2.592/(1.14)^5 + (3.1104 + 36.288)/(1.14)^6 = \$21.79$
BA II Plus: CF, CF0 = 0, C01 = 0, F01 = 2, C02 = 1.80, C03 = 2.16, C04 = 2.592, C05 = 39.3984, then NPV, I = 14, CPT = 21.79

**10. d**
Stage 1 = $[2.50 \times 1.18 / (0.14 - 0.18)] \times [1 - (1.18/1.14)^6]$ = (−73.75)(−0.22988) = $16.95 (both factors are negative, so the result is positive)
$D_6 = 2.50 \times (1.18)^6 = 6.7489$
$P_6 = 6.7489 \times 1.05 / (0.14 - 0.05) = \$78.74$
$P_0 = 16.95 + 78.74 / (1.14)^6 = 16.95 + 35.87 = \$52.83$
BA II Plus: 1.14 ÷ 1.18 − 1 = −0.033898 (I/Y = −3.3898)
BA II Plus: 6 N, −3.3898 I/Y, −2.50 PMT, 0 FV, CPT PV = 16.95
BA II Plus: 6 N, 14 I/Y, 0 PMT, −78.74 FV, CPT PV = 35.87

**11. d**
$D_t = 1.00 \times (1.12)^t$, and g = R, so every dividend has PV = $1.00
**PV of dividends** = 5 × 1.00 = $5.00
$\mathbf{EPS_5} = 3.00 \times (1.10)^5 = 4.8315$
$\mathbf{EPS_6} = 4.8315 \times 1.10 = 5.3147$
$\mathbf{P_5}$ = 14 × 5.3147 = $74.41 (forward P/E is applied to year-6 EPS)
**PV of** $\mathbf{P_5} = 74.41 / (1.12)^5 = \$42.22$
$\mathbf{P_0}$ = 5.00 + 42.22 = $47.22
**BA II Plus:** CF, CF0 = 0, C01 = 1.12, C02 = 1.2544, C03 = 1.404928, C04 = 1.573519, C05 = 76.1679, then NPV, I = 12, CPT = 47.22

# Investment Analysis Midterm - Short Answer Quiz

1. **[Ch.2]** What defines a money market instrument? Explain why a Treasury note is not one, and name two money market instruments.

2. **[Ch.2]** In a repurchase agreement (repo), what does the dealer do, and why is a repo considered a very safe short-term loan?

3. **[Ch.2]** What are Eurodollars? Does the prefix "Euro" mean that the deposit must be held in Europe?

4. **[Ch.2]** What is a mortgage-backed pass-through security, and how does securitization shift risk?

5. **[Ch.2]** Which two characteristics define common stock as a claim on the firm, and what does each mean?

6. **[Ch.2]** Why is preferred stock called a hybrid security? What happens if the firm skips a preferred dividend?

7. **[Ch.2]** What is an American Depositary Receipt (ADR), and what is its advantage for a U.S. investor?

8. **[Ch.2]** A stock in a price-weighted index splits 2-for-1. Why must the divisor be changed, and what happens to that stock's weight in the index?

9. **[Ch.2]** What does an equally weighted index imply for an investor's portfolio, and why does maintaining it require trading?

10. **[Ch.2]** What rights does the holder of a call option have, what is the holder's maximum loss, and what happens if the asset price ends below the exercise price?

11. **[Ch.2]** When is a put option in the money, and why is being in the money not the same as being profitable for the buyer?

12. **[Ch.2]** Distinguish a long futures position from a long call position in terms of right versus obligation, up-front payment and potential loss.

13. **[Ch.2]** At what point is an investor indifferent between a taxable bond and a municipal bond, and why are municipal bonds mostly held by high-bracket investors?

14. **[Ch.2]** State the two flaws of the bank discount yield as a measure of the return on a Treasury bill, and how the bond equivalent yield corrects them.

15. **[Ch.2]** For the same Treasury bill, rank the bank discount yield, the bond equivalent yield and the effective annual yield from lowest to highest, and explain what drives each step.

16. **[Ch.2]** What does it mean for a money market fund to "break the buck," and what triggered it for the Reserve Primary Fund in 2008?

17. **[Ch.2]** How do TIPS protect an investor against inflation?

18. **[Ch.2]** Distinguish general obligation bonds from revenue bonds. Which is riskier in terms of default?

19. **[Ch.2]** Compare price-weighted, value-weighted and equally weighted indexes. Which kind of stock dominates the first two?

20. **[Ch.3]** In which market do the proceeds of a sale go to the issuing firm, and why does the secondary market still matter to firms?

21. **[Ch.3]** Distinguish firm commitment from best efforts underwriting. Who bears the risk of an unsold issue in each?

22. **[Ch.3]** What does the evidence say about average first-day and long-run IPO returns, and what explains the first-day pattern?

23. **[Ch.3]** A dealer quotes a bid of $34.75 and an asked price of $34.90. At which price does an investor buy, at which does she sell, and what does the spread represent?

24. **[Ch.3]** Compare a market order and a limit order in terms of certainty of execution and certainty of price.

25. **[Ch.3]** A stock trades at $50 and an investor places a limit buy order at $47. What happens to the order, and how does it relate to market liquidity?

26. **[Ch.3]** What does a stop-loss sell order at $55 do for an investor who holds a stock trading at $60, and what does it not guarantee?

27. **[Ch.3]** What is high-frequency trading, and what do proponents and critics argue about its effect on liquidity?

28. **[Ch.3]** Explain how buying on margin affects an investor's return and risk, and who sets the initial margin requirement.

29. **[Ch.3]** When does a margin call occur for a margin-purchased (long) position and for a short sale, and why do the two differ?

30. **[Ch.3]** Describe a short sale: what does the seller do, when does the position profit, and how do the maximum gain and loss compare?

31. **[Ch.3]** In a short sale, what happens to the sale proceeds, what margin is required, and who receives the dividends paid on the borrowed shares?

32. **[Ch.3]** How do ECNs and algorithmic (high-frequency) trading change the way securities are traded, and what are the benefits of electronic trading?

33. **[Ch.3]** Name the four pillars of securities regulation and state what each does.

34. **[Ch.4]** What exactly does an investor who owns 2% of a mutual fund's shares own, and what is not guaranteed?

35. **[Ch.4]** Name the four core services an investment company provides to small investors, and say whether a return above the market is one of them.

36. **[Ch.4]** How is a fund's net asset value (NAV) computed, which items are deducted, and why does issuing new shares at NAV leave NAV unchanged?

37. **[Ch.4]** At what price is an order to buy open-end mutual fund shares at 10:00 a.m. filled, and how do closed-end funds and ETFs differ?

38. **[Ch.4]** Describe a closed-end fund: how its shares are issued, who trades with whom, and how its market price relates to NAV.

39. **[Ch.4]** What is a unit investment trust (UIT), and why are its management fees very low?

40. **[Ch.4]** How does a hedge fund differ from a mutual fund in who may invest, regulation, strategies, fees and redemptions?

41. **[Ch.4]** Distinguish a balanced fund from an asset allocation fund.

42. **[Ch.4]** Why do investors tend to underestimate a fund's expense ratio, and is the reported fund return before or after it?

43. **[Ch.4]** What is a 12b-1 fee, how large can it be, and why is it economically like a sales load?

44. **[Ch.4]** What does a back-end load (CDSC) that declines each year do economically, and who receives it?

45. **[Ch.4]** A fund reports an 11% one-year return and an investor paid a 5% front-end load. Does the reported return equal her return?

46. **[Ch.4]** Class A shares charge a front-end load and a lower expense ratio; Class C shares charge no front-end load and a higher expense ratio. Which investor should prefer which class, and why?

47. **[Ch.4]** An ETF trades above the value of its underlying basket. Describe what an authorized participant does and how this restores the ETF price.

48. **[Ch.4]** Why do ETF holders generally avoid the capital gains distributions that mutual fund holders receive when other investors redeem?

49. **[Ch.4]** What does a 2× leveraged ETF deliver, and why can its multi-period return differ widely from twice the index return?

50. **[Ch.4]** Where does the high distribution of a covered-call ETF come from, and what does the fund give up in return?

51. **[Ch.4]** Why must the average actively managed dollar underperform the market after costs? Use the logic of passive and active investors.

52. **[Ch.4]** Of 164 U.S. large-cap funds in the top quartile in 2021, none were still in the top quartile in 2023, while chance alone would give 6.25%. What does this suggest about choosing funds by past returns?

53. **[Ch.4]** A fund charges a 5.75% front-end load and you invest $10,000. Why must the fund earn more than 5.75% just for you to break even?

54. **[Ch.5]** Define the holding-period return (HPR), split it into its two components, and state which price is the denominator.

55. **[Ch.5]** Distinguish the arithmetic average from the geometric average of returns. Which one is the actual compound growth rate, and which is larger?

56. **[Ch.5]** How are the dollar-weighted and the time-weighted returns calculated, and when do they differ? Which one is better for judging a manager?

57. **[Ch.5]** For a given APR, how does the compounding frequency affect the effective annual rate (EAR), and how is a continuously compounded rate related to the EAR?

58. **[Ch.5]** Why does the shortcut real return ≈ nominal − inflation overstate the real return, and what is the exact relation?

59. **[Ch.5]** A bank reports a one-day 99% VaR of ₩40 million. What does the number mean, how often should it be exceeded, and is it the maximum possible loss?

60. **[Ch.5]** Under normally distributed returns, how is a one-day VaR scaled to a T-day horizon, and which common mistake does the rule avoid?

61. **[Ch.5]** As an investor changes the fraction y invested in the risky portfolio P, what happens to the Sharpe ratio of the complete portfolio, and why?

62. **[Ch.5]** How does the nominal interest rate respond to an increase in expected inflation, and what does this preserve?

63. **[Ch.5]** Explain what skew and kurtosis measure, and why they matter when returns are not normal.

64. **[Ch.5]** Compare historical simulation and Monte Carlo simulation as ways to obtain the return distribution for VaR.

65. **[Ch.5]** Why is Expected Shortfall (ES) used alongside VaR, and how did Basel treat the two?

66. **[Ch.5]** What does the Sharpe ratio measure, why is it unchanged when the investor mixes in the risk-free asset, and why is it the wrong tool for a single security?

67. **[Ch.5]** In the utility function $U = E(r) - \tfrac{1}{2}A\sigma^2$, what does A measure, and how do risk-averse, risk-neutral and risk-loving investors differ? What is a certainty equivalent?

68. **[Ch.5]** In $y^* = [E(r_p) - r_f] \div (A\sigma_P^2)$, how does the optimal fraction in the risky portfolio respond to the risk premium, the risk aversion A and the variance?

69. **[Ch.5]** Why does the capital allocation line bend (kink) at the risky portfolio P when the investor can borrow only at a rate above the risk-free rate?

70. **[Ch.5]** What is the capital market line, how is passive investing defended, and what is the Grossman-Stiglitz paradox?

71. **[Ch.6]** Distinguish systematic risk from firm-specific risk. Which can be diversified away, and which earns a risk premium?

72. **[Ch.6]** Suppose every stock has the same standard deviation and every pair has a correlation of 0.30. What happens to an equally weighted portfolio's standard deviation as n grows without limit, and what does the floor represent?

73. **[Ch.6]** In the two-asset portfolio variance formula, which term is the only one that can be negative, and why does it matter for diversification?

74. **[Ch.6]** When can two risky assets be combined into a zero-risk portfolio, and what determines the weights?

75. **[Ch.6]** Starting from 100% in a bond fund with $\sigma_B = 10\%$, when does adding a little of a riskier stock fund with $\sigma_S = 20\%$ reduce portfolio risk?

76. **[Ch.6]** Why is the minimum-variance portfolio generally not the best risky portfolio to combine with T-bills, and which portfolio is chosen instead?

77. **[Ch.6]** What does the separation property imply about the optimal risky portfolio and about the role of risk aversion? When does it break down?

78. **[Ch.6]** When does an investor hold more than 100% of her wealth in the optimal risky portfolio, and what does that mean for her financing?

79. **[Ch.6]** Why is a full Markowitz optimization hard to implement for many securities, and what must be estimated?

80. **[Ch.6]** State the insurance principle. Why is holding mostly your employer's stock in a 401(k) plan the opposite of it?

81. **[Ch.6]** Define covariance and the correlation coefficient. Why is the correlation easier to interpret, and what values can it take?

82. **[Ch.6]** In the 50/50 portfolio of a stock fund and a bond fund (standard deviations 21.73% and 3.54%), in what sense is the benefit of diversification "cost-free"?

83. **[Ch.6]** In which single case is there no benefit from diversification, and what does the opportunity set look like?

84. **[Ch.6]** On the opportunity set of two risky assets, which portfolios are efficient and which are dominated? Why?

85. **[Ch.6]** Define the efficient frontier of risky assets and how it is found. Where do the individual assets lie relative to it?

86. **[Ch.6]** Describe the four-step recipe for building the optimal complete portfolio. Which step depends on who the investor is?

87. **[Ch.6]** What is home bias, why does investing abroad lower portfolio risk, and what is the honest caveat?

## Answers

**1. [Ch.2]** Money market instruments are **short-term** (one year or less), **highly marketable** debt. A **Treasury note** has a maturity of 1 to 10 years, so it belongs to the capital market; negotiable **certificates of deposit**, banker's acceptances and commercial paper are money market instruments.
**Keywords: short-term · highly marketable · Treasury note · certificates of deposit**

**2. [Ch.2]** The dealer **sells government securities** to an investor and agrees to **buy them back** the next day at a slightly **higher price**; the difference is the interest. Because the securities serve as **collateral**, a repo is effectively a very safe **short-term** loan to the dealer.
**Keywords: sells securities · buy back · higher price · short-term · collateral**

**3. [Ch.2]** Eurodollars are **dollar-denominated** deposits held at **foreign banks or foreign branches of U.S. banks**, outside U.S. bank regulation. The prefix refers to the **location** of the deposit, **not necessarily Europe**: a dollar deposit in a Tokyo bank is a Eurodollar deposit.
**Keywords: dollar-denominated · foreign banks · location**

**4. [Ch.2]** A pass-through is an **ownership claim** on a **pool of mortgages**; the homeowners' monthly **principal and interest** are **passed through** to investors net of servicing fees. Securitization shifts **default and prepayment risk** from the originating banks to the investors.
**Keywords: ownership claim · pool of mortgages · principal and interest · passed through · default and prepayment**

**5. [Ch.2]** Common stock is a **residual claim**: shareholders are **paid last**, after employees, suppliers and bondholders. It carries **limited liability**: the most a shareholder can lose is the **amount invested**.
**Keywords: residual claim · paid last · limited liability · amount invested**

**6. [Ch.2]** Preferred stock is a **hybrid**: like a bond it pays a **fixed dividend** and normally has **no voting rights**, but like equity the dividend is **not a legal obligation**, so skipping it does not trigger default; **cumulative** preferred must clear all arrears before any common dividend.
**Keywords: hybrid · fixed dividend · not a legal obligation · no voting · cumulative**

**7. [Ch.2]** An **ADR** is a **dollar-denominated certificate** issued by a **depositary bank** that represents ownership of **foreign shares** and trades on a **U.S. exchange**, so U.S. investors trade and receive dividends in dollars without dealing in a foreign market or currency.
**Keywords: certificate · foreign shares · U.S. exchange · dollar-denominated · depositary bank**

**8. [Ch.2]** A split lowers the stock's price, which would mechanically lower the index, so the **divisor** is reset to keep the **index level unchanged**. The split stock's **weight falls permanently**, a known distortion of price weighting.
**Keywords: divisor · index level unchanged · weight falls · permanent**

**9. [Ch.2]** It corresponds to investing an **equal dollar** amount in each stock. As prices move the weights drift, so it needs periodic **rebalancing**: **selling outperformers** and **buying underperformers**. This implies **trading costs** that value-weighted indexes do not have.
**Keywords: equal dollar · rebalancing · selling outperformers · buying underperformers · trading costs**

**10. [Ch.2]** The holder has the **right to buy** at the **exercise price**, with no obligation. If the asset ends below that price the option **expires worthless**; the **maximum loss** is the **premium** paid, while the gain is open-ended. Unlimited loss describes the call writer, not the holder.
**Keywords: right to buy · exercise price · expires worthless · premium · maximum loss**

**11. [Ch.2]** A put gives the **right to sell** at the exercise price, so it is **in the money** when the asset price is **below the exercise price**. The buyer profits only if the price falls far enough to **cover the premium** paid; the writer's gain is capped at that premium.
**Keywords: right to sell · in the money · below the exercise price · premium · cover the premium**

**12. [Ch.2]** A long **futures** position is an **obligation** to take delivery (or settle) at the agreed price, so **losses** can be large if the price falls; a long **call** is a **right** that the holder abandons if unprofitable. A call requires a **premium** at purchase, while futures require no premium, only a **margin** deposit that acts as a performance bond.
**Keywords: obligation · right · premium · margin · losses**

**13. [Ch.2]** The investor is **indifferent** when the **after-tax yields are equal**, $r_{taxable}(1 - t) = r_{muni}$, which defines a critical **marginal tax rate** t. Investors in **higher brackets** prefer the **tax-exempt** municipal bond; those below that rate prefer the taxable bond.
**Keywords: indifferent · after-tax yields are equal · marginal tax rate · higher bracket · tax-exempt**

**14. [Ch.2]** The **bank discount yield** has two flaws: "First, it assumes that the year has only **360 days**. Second, it computes the yield as a fraction of **face value** rather than of the **price the investor paid**." The **bond equivalent yield** corrects both: it uses a **365**-day year and divides by the price paid.
**Keywords: bank discount yield · 360 days · face value · price the investor paid · bond equivalent yield · 365**

**15. [Ch.2]** The **bank discount yield** is the **lowest** because it divides by face value instead of the **price paid** and uses a 360-day year. The **bond equivalent yield** corrects both and is higher. The **effective annual yield** is the **highest** because it adds **compounding** of the sub-year return, so BDY < BEY < EAY.
**Keywords: bank discount yield · lowest · price paid · bond equivalent yield · effective annual yield · compounding · highest**

**16. [Ch.2]** Money market funds try to hold their **share value at $1** and pass along all earnings as interest. A fund "breaks the buck" when losses force its value per share **below $1**. After **Lehman** Brothers went bankrupt in September 2008, the Reserve Primary Fund, which held Lehman **commercial paper**, fell to $0.97, and the resulting wave of redemptions looked like a **run** on a bank.
**Keywords: share value at $1 · below $1 · Lehman · commercial paper · run**

**17. [Ch.2]** The **principal** amount of a TIPS is **adjusted** in proportion to increases in the **Consumer Price Index**, so coupons paid on that principal rise with inflation. TIPS therefore provide a constant stream of income in **real** (inflation-adjusted) dollars.
**Keywords: principal · adjusted · Consumer Price Index · real**

**18. [Ch.2]** **General obligation bonds** are backed by the **full faith and credit** (the taxing power) of the issuer, while **revenue bonds** are issued to finance **particular projects** (airports, hospitals, turnpikes) and are backed by the **revenues from that project** or the agency operating it. Revenue bonds are **riskier** in terms of default.
**Keywords: general obligation · full faith and credit · revenue bond · particular projects · revenues from that project · riskier**

**19. [Ch.2]** A **price-weighted** index averages share prices, so a **high-priced** stock dominates it. A **value-weighted** index weights each stock by its **market capitalization**, so a **large-cap** firm dominates it. An **equally weighted** index is the simple average of each stock's return.
**Keywords: price-weighted · high-priced · value-weighted · large-cap · equally weighted**

**20. [Ch.3]** In the **primary market** a firm sells **newly issued** securities (for example an IPO) and receives the **proceeds**. Later trades between investors take place in the **secondary market** and raise no new capital, but the **liquidity** and prices it provides set the terms on which new issues can be sold.
**Keywords: primary market · newly issued · proceeds · secondary market · liquidity**

**21. [Ch.3]** In a **firm commitment** the syndicate **buys the entire issue** at a negotiated price and resells it; the spread is its pay and the **underwriter bears the risk** of a poor sale. In **best efforts** the bank only acts as an **agent** and unsold shares revert to the issuer.
**Keywords: firm commitment · buys the entire issue · underwriter bears · best efforts · agent**

**22. [Ch.3]** IPOs are on average **underpriced**: **first-day** returns are large and positive, **money left on the table** by the issuer. Explanations include compensating institutions for revealing demand in **book-building** and the **winner's curse**. Long-run returns reverse, as IPOs tend to **underperform** comparable non-issuing firms.
**Keywords: underpriced · first-day · money left on the table · book-building · underperform**

**23. [Ch.3]** The investor **buys at the asked** price ($34.90) and **sells at the bid** ($34.75). The **spread** of $0.15 is the dealer's compensation for **immediacy** and a **round-trip** transaction cost for the investor on top of commissions.
**Keywords: buys at the asked · sells at the bid · spread · immediacy · round-trip**

**24. [Ch.3]** A **market order executes immediately** at the **best available price**, so execution is certain but the price is not. A **limit order** names the **worst acceptable price**, so the price is controlled, but it **may never execute** and waits in the limit order book.
**Keywords: market order · executes immediately · best available price · limit order · worst acceptable price · may never execute**

**25. [Ch.3]** The order **cannot be filled** now; it sits in the **limit order book** and executes only if the price **falls to $47 or below** (it may expire unfilled). Such standing orders **supply liquidity** to investors who submit market orders.
**Keywords: cannot be filled · limit order book · falls to $47 · supply liquidity**

**26. [Ch.3]** The order lies **dormant** until the stock trades **at or below $55**, then becomes a **market order** to sell, which helps **limit further losses**. It does **not guarantee** the price: in a fast-falling or gapping market the sale can be filled well below $55.
**Keywords: dormant · market order · at or below $55 · limit further losses · not guarantee**

**27. [Ch.3]** High-frequency trading is **computerized** trading that exploits very **small, fleeting price discrepancies**, with positions held for **milliseconds**. Proponents say it **narrows spreads and adds liquidity**; critics note that the liquidity can be **withdrawn** under stress, as in the 2010 flash crash.
**Keywords: computerized · small · milliseconds · narrows spreads · withdrawn**

**28. [Ch.3]** Margin **magnifies both gains and losses** on the investor's equity because the loan is fixed while the collateral value moves. **Interest accrues** whatever the stock does, so the stock must earn more than the **borrowing rate** for leverage to pay. The initial margin requirement is set by the **Federal Reserve**, and brokers may be stricter.
**Keywords: magnifies · gains and losses · interest accrues · borrowing rate · Federal Reserve**

**29. [Ch.3]** For a **long** margin purchase the loan is fixed, so a **price fall** cuts equity below the **maintenance margin**; for a **short** position the liability grows when the **price rises**, so the call comes on a rise. In both cases the broker demands **additional funds** or sells shares.
**Keywords: long · price fall · short · price rises · maintenance margin · additional funds**

**30. [Ch.3]** The short seller **borrows** shares and **sells** them, then **buys back** later to return them, profiting if the **price falls**. The gain is **capped** because the price cannot fall below zero, while the loss is **unlimited** because the price can rise without bound.
**Keywords: borrows · sells · buys back · price falls · capped · unlimited**

**31. [Ch.3]** The sale **proceeds stay in the account as collateral**, and the short seller must post **additional margin**. The lender keeps the economic rights of ownership, so the short seller must **pay the lender any dividends** the stock pays, a **carrying cost** of the trade.
**Keywords: collateral · additional margin · dividends · pay the lender · carrying cost**

**32. [Ch.3] ECNs** (electronic communication networks) match buy and sell orders directly with **no dealer** in the middle. **Algorithms** decide the timing, price and size of orders to cut cost and market impact, and ultra-fast high-frequency programs exploit tiny price gaps in **milliseconds**. The benefits are faster execution, **lower trading costs**, greater price transparency, anonymity for large orders and trading beyond exchange hours.
**Keywords: ECN · no dealer · algorithm · milliseconds · lower trading costs**

**33. [Ch.3]** The **Securities Acts of 1933** and 1934 require **full disclosure** at issue and created the **SEC** to oversee markets. **Self-regulation (FINRA)** means industry bodies police member conduct and trading practices. **Sarbanes-Oxley** (2002) tightened corporate governance, auditing and financial reporting after major scandals. **Insider-trading** rules prohibit trading on **material non-public information**.
**Keywords: 1933 · full disclosure · SEC · FINRA · Sarbanes-Oxley · material non-public information**

**34. [Ch.4]** She owns a **proportional (pro-rata) claim** on the **whole pooled portfolio**, 2% of every asset, and receives 2% of its **income**. She does **not own specific securities**, and no return is **guaranteed**.
**Keywords: proportional · whole pooled portfolio · income · specific securities**

**35. [Ch.4]** The four services are **record keeping and administration**, **diversification and divisibility**, **professional management**, and lower **transaction costs** from trading in large blocks. A **guaranteed** return above the market is not among them; whether management beats the market is an empirical question.
**Keywords: record keeping · diversification · professional management · transaction costs · guaranteed**

**36. [Ch.4] NAV** = (**market value of assets** − **liabilities**, such as accrued management fees) ÷ **shares outstanding**. New shares are issued **at NAV**, so assets and shares rise in **proportion** and NAV is unchanged, with **no dilution** of existing holders.
**Keywords: market value of assets · liabilities · shares outstanding · at NAV · proportion**

**37. [Ch.4]** It is filled at the **NAV struck once a day at the 4:00 p.m. close**, the **same price** for every order that day, and the investor does **not know** it when ordering. Closed-end funds and ETFs trade on an **exchange** at intraday market prices.
**Keywords: NAV · 4:00 p.m. · same price · not know · exchange**

**38. [Ch.4]** A closed-end fund has a **fixed number** of shares after its **IPO** and neither issues nor redeems them. Investors trade with each other on an **exchange**, so the price is set by **supply and demand** and can sit at a **premium or discount** to NAV.
**Keywords: fixed number · IPO · exchange · supply and demand · premium or discount**

**39. [Ch.4]** A UIT is a **fixed, unmanaged** portfolio deposited in a trust by a **sponsor** and sold as redeemable units; it **terminates on a set date**. With essentially **no trading** after formation, **management fees are very low**.
**Keywords: fixed · terminates · sponsor · no trading · very low**

**40. [Ch.4]** Hedge funds are private pools open only to **accredited and institutional** investors and largely **exempt from the Investment Company Act** of 1940. They use **short sales, derivatives and heavy leverage**, charge higher fees (**2 and 20**), and restrict redemptions through **lock-ups**.
**Keywords: accredited · Investment Company Act · short sales · 2 and 20 · lock-up**

**41. [Ch.4]** A **balanced fund** holds **stocks and bonds** in a relatively **stable mix**; an **asset allocation fund** actively **shifts** the mix across asset classes as market conditions change.
**Keywords: balanced fund · stocks and bonds · stable mix · asset allocation fund · shifts**

**42. [Ch.4]** Operating expenses are **deducted daily from fund assets** before NAV is struck, so the investor is **never billed** separately and the cost is easy to overlook. The reported return is already **net of** the expense ratio.
**Keywords: deducted · daily · from fund assets · never billed · net of**

**43. [Ch.4]** A **12b-1 fee** is an **annual** charge of up to **1%** of assets that pays for **marketing and distribution** (including adviser compensation). It is reported inside the **expense ratio**, but economically it is a **load** paid in instalments.
**Keywords: annual · marketing and distribution · 1% · expense ratio · load**

**44. [Ch.4]** A **CDSC** is charged on redemption and **declines** each year until it disappears. It makes **early redemption** expensive and so **locks investors in**; it compensates the **selling broker**, not the fund.
**Keywords: CDSC · declines · early redemption · locks investors in · selling broker**

**45. [Ch.4]** No: the reported return **overstates** her return. Published returns are computed from **NAV**, which is net of the **expense ratio** but **not of loads**; the load **reduces the amount invested**, so over one year a 5% load can consume almost half of an 11% return.
**Keywords: overstates · not of loads · NAV · expense ratio · reduces the amount invested**

**46. [Ch.4]** The load is a **one-time** cost but the higher expense ratio is paid **every year** and compounds. Over a **short** horizon the load dominates, so **Class C** wins; over a **long** horizon the annual fee dominates, so **Class A** wins.
**Keywords: Class A · Class C · long · short · one-time · every year**

**47. [Ch.4]** The **authorized participant buys the cheaper basket**, swaps it **in kind** for newly **created** ETF shares, and **sells the ETF shares** on the exchange. The selling pressure pushes the ETF price **back toward the basket value**.
**Keywords: authorized participant · buys the cheaper basket · in kind · sells the ETF shares · back toward the basket value**

**48. [Ch.4]** In a **mutual fund**, **redemptions force** the fund to **sell stock**, realizing gains that are distributed to all holders. In an **ETF** the seller finds a buyer on the **exchange**, and creations and redemptions happen **in kind**, not for cash, so the fund does not need to sell.
**Keywords: mutual fund · redemptions force · sell stock · ETF · exchange · in kind**

**49. [Ch.4]** It aims for **twice the index's daily return**; the fund **resets** to 2× every day, so gains and losses **compound** on a changing base. In **volatile** markets it falls behind even if the index ends flat, so multi-period returns are not 2×.
**Keywords: daily · twice · resets · compound · volatile**

**50. [Ch.4]** The fund **sells call options** on shares it owns and distributes the **premium** received. That premium is the price of the **upside it gives away** (gains are capped above the strike) while the **downside remains open**; it is an option strategy, **not a dividend** and not insurance.
**Keywords: sells call options · premium · upside · downside · not a dividend**

**51. [Ch.4] Passive** investors hold the market, so what remains for **active investors as a group** is also the market: their value-weighted average **gross return** equals the market return. After **fees** the average active dollar must **lag** the market. This says nothing about any single fund.

**52. [Ch.4]** Past top performance is **not a reliable** signal of future top performance: if **skill** drove the rankings, winners would repeat more often than **chance**, not less. The evidence shows no **persistence**; the lasting signal is the **fee**, not past returns.
**Keywords: persistence · not a reliable · chance · skill · fee**

**53. [Ch.4]** A front-end **load** is deducted before investing, so a 5.75% load on $10,000 leaves only **$9,425** as the **net** amount invested. The load is 5.75% of the gross, but to get back to $10,000 you must earn **6.10%** on the net, i.e. you need more than the load percentage just to **break even**.
**Keywords: load · net · 9,425 · 6.10% · break even**

**54. [Ch.5] HPR** = (ending price − beginning price + dividends) ÷ beginning price, the **capital gain yield** plus the **dividend yield**. The denominator is always the **purchase price**, **never the ending price**.
**Keywords: capital gain yield · dividend yield · purchase price · never the ending price**

**55. [Ch.5]** The **arithmetic** average is the simple mean of period returns; the **geometric** average is the constant **compound** growth rate that reproduces the actual ending wealth. The geometric average is **smaller** than the arithmetic one when returns fluctuate, by roughly $\frac{1}{2}\sigma^2$, and the gap grows with **volatility**.
**Keywords: arithmetic · geometric · compound · smaller · volatility**

**56. [Ch.5]** The **dollar-weighted** return is the **IRR** of all **cash flows** (purchases, dividends, sales); the time-weighted return is the **geometric** average of **sub-period** returns. They **differ** when money is added or withdrawn at different times, and the time-weighted figure ignores that timing, so it is better for judging the **manager**.
**Keywords: IRR · cash flows · geometric · differ · manager**

**57. [Ch.5]** For a given APR, **more frequent** compounding gives a **higher EAR**, $\text{EAR} = (1 + \text{APR}/m)^m - 1$. With **continuous** compounding the APR equals the **natural log** of (1 + EAR).
**Keywords: EAR · more frequent · higher · continuous · natural log**

**58. [Ch.5]** The exact relation is the **Fisher** equation, $1 + r = (1 + R) \div (1 + i)$. The shortcut $R - i$ is an **approximation** that **overstates** the real return by the **cross term** $r \times i$, and the error grows when inflation is high.
**Keywords: Fisher · approximation · overstates · cross term**

**59. [Ch.5] VaR** is the loss that is **exceeded** with a given **probability** over the horizon; here the loss should exceed ₩40 million on about **1 trading day in 100**. It is **not the maximum** loss: on the worst days losses can be **larger**, and VaR says nothing about how much larger.
**Keywords: probability · exceeded · 1 trading day in 100 · not the maximum · larger**

**60. [Ch.5]** Under normal returns $\mathbf{VaR = V_0 \times z \times \sigma \times \sqrt{T}}$: the one-day figure is scaled by the **square root** of time because independent daily variances add, **not multiplied by T**, which would overstate the risk. The **z-value** reflects the **confidence level** (1.645 for 95%, 2.326 for 99%) and $\sigma$ is the daily **volatility**.
**Keywords: square root · not multiplied by T · z-value · volatility**

**61. [Ch.5]** It is **unchanged** for every y: both the **risk premium** and the **standard deviation** of the complete portfolio **scale** with y, so their ratio equals the **slope** of the capital allocation line. This is why the choice of P and the choice of y can be solved separately.
**Keywords: unchanged · risk premium · standard deviation · scale · slope**

**62. [Ch.5]** Investors need compensation for expected inflation, so the **nominal** rate **rises** with **expected inflation**: $r_{nom} = r_{real} + E(i)$. The increase in the nominal rate is needed to **preserve** investors' **real** rate of return.
**Keywords: expected inflation · nominal · real · rises · preserve**

**63. [Ch.5] Skew** measures the **asymmetry** of the return distribution. **Kurtosis** compares the frequency of **extreme** values with that of the **normal distribution** (whose kurtosis is zero), so it is called a measure of **fat tails**, which signal tail risk that the standard deviation alone understates.
**Keywords: skew · asymmetr · kurtosis · fat tails · extreme · normal distribution**

**64. [Ch.5] Historical simulation** sorts the **actual past returns** and reads off the quantile; **no distribution** is assumed so **fat tails** come for free, but it needs a long sample. **Monte Carlo** simulation draws many paths from an **assumed model**, handles **options** and other non-linear payoffs, but carries model risk.
**Keywords: historical simulation · actual past returns · no distribution · fat tails · Monte Carlo · assumed model · options**

**65. [Ch.5]** VaR marks the **cut-off** but says nothing about how bad the loss is beyond it. ES (also CVaR) is the **average loss** in the tail **beyond** VaR. ES is **coherent** (subadditive) where VaR is not, so merging books cannot manufacture risk, and **Basel FRTB** replaced the 99% VaR capital charge with **97.5%** expected shortfall.
**Keywords: cut-off · average loss · beyond · coherent · Basel · 97.5%**

**66. [Ch.5]** The Sharpe ratio is the **reward per unit** of volatility, $[E(r_P) - r_f] / \sigma_P$. It equals the **slope of the CAL** and is **unchanged** however much of the risk-free asset is mixed in, so it measures the quality of the risky portfolio itself. It is the wrong tool for a **single security** because the denominator includes **diversifiable risk**.
**Keywords: reward per unit · slope of the CAL · unchanged · single security · diversifiable risk**

**67. [Ch.5]** A **risk-averse** investor has $A > 0$, a **risk-neutral** investor has $A = 0$ and judges by E(r) alone, and a **risk-loving** investor has $A < 0$. The term $\frac{1}{2}A\sigma^2$ is the **penalty** for risk. The **certainty equivalent** is the rate that a risk-free investment would have to offer to give the same utility as the risky portfolio.
**Keywords: risk-averse · risk-neutral · risk-loving · penalty · certainty equivalent**

**68. [Ch.5]** $y^*$ **maximizes utility** and **rises with the risk premium**, **falls with risk aversion** and **falls with variance**: a larger reward per unit of risk raises the risky share, while a larger risk penalty lowers it.
**Keywords: maximizes utility · rises with the risk premium · falls with risk aversion · falls with variance**

**69. [Ch.5]** To the left of P the investor **lends** at the risk-free rate. Beyond P she must borrow at a rate above $r_f$, and because the **borrowing rate is higher** the **slope falls** (the Sharpe ratio of leveraged positions is lower), so the CAL has a **kink** at P.
**Keywords: lends · borrowing rate is higher · slope falls · kink**

**70. [Ch.5]** When a **broad market index** is used as the risky portfolio P, the resulting capital allocation line is the **capital market line** (CML). Passive investing is defended by the **free-rider** argument (the analysis of thousands of other investors is already in the price) and by **low cost** (index funds charge about 0.03-0.2% a year against 1% or more for active funds). The open question is **who sets prices** if everyone is passive, the **Grossman-Stiglitz** paradox.
**Keywords: capital market line · broad market index · free-rider · low cost · who sets prices · Grossman**

**71. [Ch.6]** Total variance is **systematic** plus **firm-specific** variance. Adding stocks shrinks **firm-specific** (diversifiable) risk toward zero but leaves systematic (market) risk. Prices are set by **diversified investors**, so only the systematic part earns a **risk premium**.
**Keywords: systematic · firm-specific · diversified investors · risk premium**

**72. [Ch.6]** As n grows the firm-specific term $(1/n)\sigma^2$ vanishes and $\sigma_P$ **falls toward a floor** equal to the square root of the **average covariance**. The floor is **market (systematic) risk**, which **cannot be diversified** away.
**Keywords: falls · floor · average covariance · market risk · cannot be diversified**

**73. [Ch.6]** The **interaction (covariance) term** $2w_S w_B \cdot \text{Cov}$ is the only one that can be **negative**; the other two terms are **squares** and always positive. When the **correlation** is below +1 it shrinks the variance, and for $\rho < 0$ it pulls $\sigma_P$ below the **weighted average** of the two standard deviations.
**Keywords: interaction term · negative · correlation · weighted average · squares**

**74. [Ch.6]** When the assets are **perfectly negatively correlated** ($\rho = -1$) a **zero-risk** portfolio exists: setting $\sigma_P = |w_S\sigma_S - w_B\sigma_B| = 0$ gives $w_S = \sigma_B \div (\sigma_S + \sigma_B)$, so the weights are set by the **standard deviations**.
**Keywords: perfect negative correlation · zero risk · standard deviations**

**75. [Ch.6]** Risk falls when the **correlation** is **below $\sigma_B \div \sigma_S$** (here $10 \div 20 = \mathbf{0.5}$), the point at which the minimum-variance stock weight becomes positive. A negative correlation is **not required**.
**Keywords: correlation · below $\sigma_B \div \sigma_S$ · 0.5 · not required**

**76. [Ch.6]** The minimum-variance portfolio only minimizes risk and **ignores expected return**. The optimal risky portfolio is the **tangency** point where the **capital allocation line** is steepest, so it maximizes the **Sharpe ratio**, which is **at least as high** as that of any other mix, including the minimum-variance portfolio.
**Keywords: ignores expected return · Sharpe ratio · tangency · capital allocation line · at least as high**

**77. [Ch.6] One optimal risky portfolio is the same for every investor** who shares the same inputs; only the **split** between it and **T-bills** depends on **risk aversion**. The property breaks down with **different inputs, borrowing rates above lending rates, taxes** and human capital.
**Keywords: same for every investor · risk aversion · split · T-bills · different inputs**

**78. [Ch.6]** A **risk-tolerant** (low risk aversion) investor gets $y^* = [E(r_P) - r_f] \div (A\sigma_P^2)$ **above 1**, so she **borrows** at the **risk-free rate** and invests more than her wealth in P, a **leveraged** position.
**Keywords: low risk aversion · above 1 · borrows · risk-free rate · leveraged**

**79. [Ch.6]** With n securities you must estimate **expected returns** for each, **n variances** and $n(n-1)/2$ **covariances**, a total of $n(n+3)/2$ that **grows with the square of n** (125,750 inputs for n = 500). Every estimate carries **estimation error**.
**Keywords: expected returns · n variances · covariances · square of n · estimation error**

**80. [Ch.6]** "Risk reduction by **spreading exposure** across many **independent risk sources** is called the **insurance principle**." An employee who also holds company stock in her 401(k) is **doubling down** on the fortunes of a **single firm**, since her job and her savings depend on the same firm.
**Keywords: insurance principle · spreading exposure · independent risk sources · single firm · doubling down**

**81. [Ch.6] Covariance** measures how much two returns move together, but its units are "**percent square**," so its size is hard to interpret. The **correlation coefficient** rescales it by the **product of the standard deviations**, $\rho = \text{Cov} \div (\sigma_S\sigma_B)$, so $\text{Cov} = \rho\sigma_S\sigma_B$, and it lies **between −1 and +1**.
**Keywords: covariance · percent square · correlation coefficient · product of the standard deviations · between −1 and +1**

**82. [Ch.6]** The portfolio's expected return (8.00%) is exactly the **weighted average** of the two funds' returns, so you **lost nothing** in return. Yet the portfolio standard deviation (9.57%) is far **below** the weighted average of the two (12.64%). That 3.07-point gap is the **diversification benefit**, and it exists purely because the **correlation** is below +1.
**Keywords: weighted average · lost nothing · below · correlation · diversification benefit**

**83. [Ch.6] Perfect positive correlation** ($\rho = +1$) is the **only** case with no benefit from diversification: $\sigma_P = w_B\sigma_B + w_S\sigma_S$ is a **weighted average** of the component standard deviations and the opportunity set is a **straight line** through the two assets. Whenever returns are **less than perfectly** positively correlated, $\sigma_P$ is below that weighted average.
**Keywords: perfect positive correlation · only · weighted average · straight line · less than perfectly**

**84. [Ch.6]** The **upward-sloping** part of the opportunity set, starting at the **minimum-variance portfolio**, is the **efficient** part. Points below it are **dominated**, because a portfolio on the efficient part offers a **higher expected return** at the same standard deviation.
**Keywords: efficient · upward-sloping · minimum-variance portfolio · dominated · higher expected return**

**85. [Ch.6]** The **efficient frontier** is the set of portfolios that offer the **highest expected return** for each level of risk. It is found by **minimizing variance** subject to each **target expected return**. Individual assets all lie **inside** the frontier, so every one of them is dominated.
**Keywords: efficient frontier · highest expected return · minimizing variance · target expected return · inside**

**86. [Ch.6]** Step 1 collects the **inputs**: expected returns, standard deviations and covariances or correlations. Step 2 traces the **efficient frontier** (upper branch). Step 3 **maximizes the Sharpe ratio** to find the optimal risky portfolio P*. Step 4 sets **y*** from the investor's **risk aversion** and splits wealth between P* and the risk-free asset. Only step 4 depends on the investor; steps 1-3 are identical for everyone, the **separation property**.
**Keywords: inputs · efficient frontier · Sharpe ratio · y* · separation property**

**87. [Ch.6]** Investors everywhere **overweight** their home country relative to its share of the world portfolio, a pattern called **home bias**. Adding foreign assets lowers the **covariance** term in the portfolio variance because countries' systematic shocks differ. The caveat is that global **correlations** have risen and **currency** risk adds another layer, yet the benefit is not zero as long as correlations stay **below +1**.
**Keywords: home bias · overweight · covariance · correlations · below +1 · currency**